Capital–Skill Complementarity, Consumption Upgrading, and the Rising Skill Premium

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Abstract

Using household expenditure data and a new occupation-based measure of skill intensity, I document two facts. First, higher-income households allocate a larger share of their spending to skill-intensive goods and services, suggesting that income growth can raise the skill premium through demand. Second, holding real income fixed, consumption has shifted toward these goods and services over time, pointing to additional roles for technological change and relative prices. To quantify the sources of the rise in the U.S. skill premium, I develop a model that combines capital–skill complementarity, final-good-specific production technologies, and nonhomothetic preferences. I provide new estimates of capital–skill complementarity at the final-good level, showing that equipment complements high-skill labor and substitutes for low-skill labor. Equipment accumulation accounts for 83.5% of the predicted rise in the skill premium, and productivity-driven consumption upgrading for 8.4%.

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1 Introduction

Over the past several decades, there has been a secular rise in the wage gap between high- and low-skill workers. A large literature attributes this increase to skill-biased technological change, with capital–skill complementarity providing an important mechanism through which equipment accumulation raises the relative demand for skilled labor (e.g. Krusell et al. (2000), Autor et al. (1998), Acemoglu and Autor (2011)). At the same time, economic growth has substantially changed the composition of household consumption, potentially shifting demand toward skill-intensive goods and services (Leonardi (2015), Buera et al. (2022), Comin et al. (2026)). Distinguishing these mechanisms is important because similar patterns of demand reallocation can arise from technological change or shifts in consumption, with different implications for relative prices, factor demand, and the distribution of income. This paper jointly quantifies these mechanisms and asks how much of the rise in the skill premium is driven by equipment accumulation through capital–skill complementarity and how much is attributable to consumption upgrading induced by productivity growth.

To separate these sources, I develop a unified framework that combines capital–skill complementarity, industry-specific technologies, and nonhomothetic preferences. Equipment accumulation raises relative skill demand within industries through capital–skill complementarity, while productivity-driven income growth reallocates expenditure toward more income-elastic goods and services. Industry-specific productivity growth further affects relative goods prices and the allocation of production and consumption across industries. Using a new occupational-requirement-based measure of high-skill labor and new estimates of the elasticities governing capital–skill complementarity, I discipline each mechanism separately. I find that equipment accumulation is the dominant force behind the rising skill premium, while consumption upgrading provides a smaller but economically meaningful amplification.

A key input into the analysis is a new measure of skill intensity based on the preparation required to perform each occupation. I use O*NET *Job Zones* to classify detailed occupations according to their education, training, and experience requirements, and combine this classification with occupation-by-industry employment and wage data from the Occupational Employment and Wage Statistics. I then use the BEA input-output tables to measure the skill intensity embodied in each final-demand category, including labor used both directly and indirectly throughout its production chain. Unlike classifications based

on workers' educational attainment or broad occupational groups, this approach distinguishes advanced positions from supporting roles within the same occupational category and allows a worker's measured skill level to evolve as the worker progresses across occupations with different preparation requirements. It also identifies a narrower group of highly skilled occupations whose aggregate employment share remains relatively stable over time. The resulting measure allows changes in the skill premium to be interpreted primarily as shifts in relative skill demand rather than as changes in the composition of the skill groups themselves.

Using this measure, I document new evidence of consumption upgrading—the shift in expenditure toward more skill-intensive goods and services as income rises—both across the income distribution and over time. At any point in time, richer households devote a larger share of their expenditures to goods and services that are relatively intensive in high-skill labor. This cross-sectional pattern provides an income channel through which economic growth can raise relative skill demand: as household income rises, expenditure shifts toward more skill-intensive goods and services, increasing the demand for skilled workers. In addition, holding real income fixed, households have increasingly shifted their expenditures toward skill-intensive goods and services over time. These time-series changes point to additional forces affecting relative skill demand through changes in relative goods prices, potentially arising from industries' differential exposure to input-price changes or uneven productivity growth across industries.

Motivated by these facts, I develop a multi-industry general equilibrium model with two types of labor, equipment capital, structures capital, and nonhomothetic demand. Production features a nested constant elasticity of substitution (CES) structure that allows equipment to be differentially substitutable with high- and low-skill labor, while industries differ in their factor intensities and productivity growth. On the demand side, industry-specific expenditure elasticities generate consumption upgrading as household income rises.

The model distinguishes three main mechanisms. First, growth in the stock of equipment capital generates skill-biased technological change through capital–skill complementarity: equipment accumulation raises the demand for high-skill labor relative to low-skill labor within industries. Second, aggregate productivity growth that is common across industries is skill neutral within each industry but can nevertheless raise the skill premium through consumption upgrading. Higher aggregate productivity raises household income and shifts expenditure toward goods and services with higher expenditure

elasticities. When these goods and services are also relatively skill intensive, this reallocation increases the economy-wide demand for high-skill labor. Third, relative industry-specific TFP growth changes relative goods prices and reallocates expenditure toward faster-productivity-growth industries. If these industries are relatively skill intensive or equipment intensive, this reallocation further raises the demand for high-skill labor and can reinforce the capital–skill complementarity channel.

One important contribution of the paper is to provide new estimates of capital–skill complementarity at the final-good level. Using the skill and equipment intensities embodied throughout each Consumer Expenditure Survey (CEX) category's production chain, I estimate the elasticities of substitution between equipment and high- and low-skill labor by exploiting differences across categories in their exposure to the secular decline in equipment prices. The estimates imply an elasticity of substitution of 1.23 between equipment and low-skill labor and an outer-nest elasticity of 0.41 between high-skill labor and the low-skill–equipment composite, providing strong evidence of capital–skill complementarity.

With the degree of capital–skill complementarity disciplined by these estimates, I calibrate the model to the U.S. economy over 1982–2019. The model predicts a 32.8% increase in the skill premium under homothetic preferences and a 38.1% increase under nonhomothetic preferences. Thus, allowing expenditure patterns to change with income amplifies the predicted increase by 5.3 percentage points. Counterfactual exercises further isolate the forces underlying this result. Equipment accumulation alone raises the skill premium by 31.8%, accounting for 83.5% of the total predicted increase, and is therefore the dominant quantitative force. Aggregate productivity growth alone has no effect on the skill premium under homothetic preferences, but raises it by 3.2% under nonhomothetic preferences as higher income shifts expenditure toward more skill-intensive goods and services. This standalone consumption-upgrading effect accounts for 8.4% of the total increase. Structures accumulation and relative industry-specific productivity growth have only modest standalone effects on relative skill demand through changes in relative production costs and goods prices, although their interactions with equipment accumulation and aggregate productivity growth provide an additional increase in the skill premium. Overall, capital–skill complementarity is the primary driver of the rising skill premium, while consumption upgrading meaningfully amplifies its increase.

My paper contributes to the broader literature on skill-biased technological change. Early studies document the importance of within-industry shifts in relative skill demand

(Katz and Murphy, 1992; Bound and Johnson, 1995; Berman et al., 1994), while subsequent work provides mechanisms based on computerization, changing task content, and capital–skill complementarity (Autor et al., 1998, 2003; Krusell et al., 2000). Building on this literature, I estimate the substitution elasticities between equipment and workers of different skill levels within a multi-industry general equilibrium framework. Equipment accumulation consequently changes relative skill demand both within industries, through capital–skill complementarity, and across industries, through differences in factor intensities.

The paper is also related to the literature on structural transformation, which distinguishes production- and demand-side sources of sectoral reallocation. On the production side, Ngai and Pissarides (2007) emphasize differences in sectoral productivity growth, while Acemoglu and Guerrieri (2008) show how capital deepening interacts with sectoral differences in factor intensities. On the demand side, nonhomothetic preferences cause income growth to reallocate consumption across sectors (Matsuyama, 1992; Echevarria, 1997; Kongsamut et al., 2001), with more recent work allowing richer income and relative-price effects (Boppart, 2014; Comin et al., 2021). Herrendorf et al. (2013) show that the relative importance of these forces depends on whether consumption is measured in final-expenditure or value-added terms. I follow the final-expenditure approach by assigning to each consumption category the labor and capital inputs used throughout its production chain. The paper that is most closely related in its production structure is Hubmer (2023), who combines nonhomothetic demand with capital–labor substitution to study the effects on labor share. I extend this framework by distinguishing high- and low-skill labor and estimating the degree of capital–skill complementarity to study the skill premium.

The consumption upgrading channel is closely related to several papers in the literature. Using cross-sectional variation in household expenditures, Leonardi (2015) documents that more educated and higher-income households spend a larger share on skill-intensive goods and services. His quantitative exercise studies a different driving force, the rise in the relative supply of skilled workers, and finds that the induced demand shift accounts for about 6.5% of the shift in relative skill demand in the United States between 1984 and 2002. Buera et al. (2022) show that productivity growth can raise the skill premium by reallocating activity toward skill-intensive sectors through both nonhomothetic preferences and sectoral-level differences in productivity growth. Comin et al. (2026) show that shocks raising the mean or dispersion of household income can generate labor-market polarization by reallocating expenditure, through nonhomothetic demand,

toward sectors intensive in high- and low-skill occupations.

My paper differs in its definition of skill and its treatment of the economy's input-output structure, but the central distinction is that I explicitly model and estimate capital–skill complementarity. These papers either exclude capital or absorb capital–skill complementarity into reduced-form changes in factor intensities and broader measures of skill-biased technological change. Industry skill intensity therefore responds endogenously to equipment accumulation, allowing the framework to separately quantify the rising skill premium due to capital accumulation and productivity growth. The decomposition identifies equipment accumulation as the dominant source of the rising skill premium and assigns a smaller role to the income-driven consumption channel than recent models that abstract from capital. Section 5.2 provides a more detailed comparison.

Finally, the paper contributes to the empirical literature estimating substitution between capital and labor. Following the insight of Griliches (1969) and the quantitative framework of Krusell et al. (2000), a large literature studies how differences in substitutability across factors shape labor-market outcomes. Related empirical work identifies substitution elasticities by exploiting heterogeneous exposure to changes in relative factor prices. Karabarbounis and Neiman (2014) use variation in investment prices and labor shares to estimate long-run capital–labor substitution, while Raval (2019) studies capital–labor substitution using more disaggregated data. My identification strategy is particularly related to Hubmer (2023), who exploits cross-good differences in equipment intensity together with the secular decline in equipment prices to identify capital–labor substitution. I extend this approach by distinguishing high- and low-skill labor and exploiting cross-industry differences in initial factor intensities together with long-run changes in equipment prices and wages to estimate the substitution elasticities between equipment capital and the two skill groups.

The remainder of the paper is organized as follows. Section 2 describes the data and presents the empirical patterns motivating the analysis. Section 3 develops the general equilibrium model and characterizes the mechanisms through which capital accumulation, productivity growth, and nonhomothetic demand affect the skill premium. Section 4 describes the estimation and calibration strategy. Section 5 presents the quantitative results and decomposition exercises, and compares the findings with closely related papers. Section 6 concludes.

2 Empirics

In this section, I describe the data used for the empirical analysis and model calibration. While the traditional approach in the literature typically defines skill groups based on educational attainment, I instead focus on differences in skill requirements across occupations. Combining occupational skill measures from O*NET with industry-level employment and wage data from OEWS, I show that the employment share of high-skill occupations remains relatively stable over time, even as the wage differential between high- and lower-skill occupations rises substantially. I then construct an industry-level measure of skill intensity based on the high-skill wage-bill share. Finally, I map this measure through the BEA I-O and the CEX to quantify the skill intensity associated with household consumption throughout the production value chain, and document a mild positive relationship between expenditure elasticities and skill intensities.

2.1 Occupation-based skill classification

O*NET. O*NET publishes quarterly information on worker characteristics for over 950 occupations within the U.S. economy since 2003.[1] The variable of interest in O*NET is an 8-digit occupation *Job Zone* classification. This classification provides a comprehensive measure of the vocational preparation required for each occupation. To determine the Job Zone for each occupation, two trained occupational experts assess various factors, including job descriptions, tasks, required levels of education, necessary work experience, on-the-job training, and the position of the occupation within a career path. They also consult the education and training classification provided by the Bureau of Labor Statistics.

In my analysis, I classify each 6-digit Standard Occupational Classification (SOC) occupation based on the aggregate skill level derived from the corresponding 8-digit SOC occupations. Occupations with $Job\ Zone \geq 4$ are classified as high skill, occupations with $2 < Job\ Zone < 4$ as middle skill, and occupations with $Job\ Zone \leq 2$ as low skill. For the model calibration in Section 4, I collapse these categories into two groups: occupations with $Job\ Zone \geq 4$ constitute the high-skill group, while those with $Job\ Zone < 4$ consti-

[1] Annual data for 1998 to 2002 are available through an extrapolated version that was evaluated and refined from existing Dictionary of Occupational Titles (DOT) data. However, researchers conducting longitudinal studies are advised to begin with the 2003 release. I use O*NET 5.1 as the baseline skill classification and incorporate classifications from more recent O*NET releases as newly introduced occupations enter the survey.

tute the low-skill group. I use $Job\ Zone = 4$ as the threshold for high-skill occupations because O*NET defines this category as requiring considerable preparation.[2] Finally, I distinguish low-skill service occupations from other low-skill occupations following the classification in Acemoglu and Autor (2011), since the literature documents substantially different long-run labor market trends for low-skill service jobs relative to other low-skill occupations.[3]

OEWS. I map the O*NET-based skill classification to occupation-level employment and wage estimates from the Occupational Employment and Wage Statistics (OEWS). OEWS is the BLS's primary establishment-based survey for measuring employment and wages at detailed occupation-by-industry levels. The modern series of occupational employment and wage estimates begins in 1997 and is based on semiannual surveys. Annual OEWS estimates combine responses from six semiannual panels collected over a three-year period. In each panel, approximately 180,000 to 200,000 establishments are sampled and stratified within states by substate area, industry, establishment size, and ownership. Across the six panels, the survey covers establishments employing roughly 83 million workers, or about 57 percent of the national workforce.

Compared with household-based surveys such as the Current Population Survey, the much larger establishment sample and the detailed occupation-by-industry employment counts in OEWS make it particularly well suited for constructing measures of skill inten-

[2]Definition for $Job\ Zone = 4$: Considerable Preparation Needed. A considerable amount of work-related skill, knowledge, or experience is needed for these occupations. For example, an accountant must complete four years of college and work for several years in accounting to be considered qualified. Most occupations in this category require a four-year bachelor's degree, although some do not. Workers in these occupations typically need several years of relevant work experience, on-the-job training, and/or vocational training. Many also involve coordinating, supervising, managing, or training others. Examples include real estate brokers, sales managers, database administrators, graphic designers, chemists, art directors, and cost estimators.

[3]Acemoglu and Autor (2011) broadly classify three groups of workers as service occupations: Protective Service, Food/Cleaning Service and Personal Care. The Census Bureau defines the service occupation as jobs that involve helping, caring for or assisting others. I follow their definition and classify workers who satisfy both of the following two requirements as low-skill service workers: first, ranked low-skill according to O*NET *Job Zone*, and second, belong to one of the four major categories according to the 2010 Standard Occupational Classification System from the BLS: 1) 33-0000: Protective Service Occupations; 2) 35-0000: Food Preparation and Serving Related Occupations; 3) 37-0000: Building and Grounds Cleaning and Maintenance Occupations; 4) 39-0000: Personal Care and Service Occupations. Remaining low-skill workers are mostly low-wage jobs that are concentrated in 43-0000: Office and Administrative Support Occupations; 45-0000: Farming, Fishing, and Forestry Occupations; 47-0000: Construction and Extraction Occupations; 49-0000: Installation, Maintenance, and Repair Occupations; 51-0000: Production Occupations; 53-0000: Transportation and Material Moving Occupations. Note that the non-service low-skill workers constitute a larger group than low-skill production workers.

sity at a highly disaggregated occupation-industry level.

Relative wages by skill group. Figure 1 plots the employment share of high-skill workers and their relative wage compared with the combined group of middle- and low-skill workers. The figure reports annual observations together with the average employment share and relative wage within each period.

Two features stand out. First, the relative wage of high-skill workers increases substantially across the two comparable periods, with the average relative wage approximately 15 percent higher in 2012–2019 than in 1997–2001. Second, the employment share of high-skill occupations changes relatively little, increasing only modestly from approximately 25 percent to 27 percent. This stability is particularly notable when compared with conventional education-based measures of skill supply: over the same period, the share of college graduates increased from approximately 20 percent to more than 35 percent.

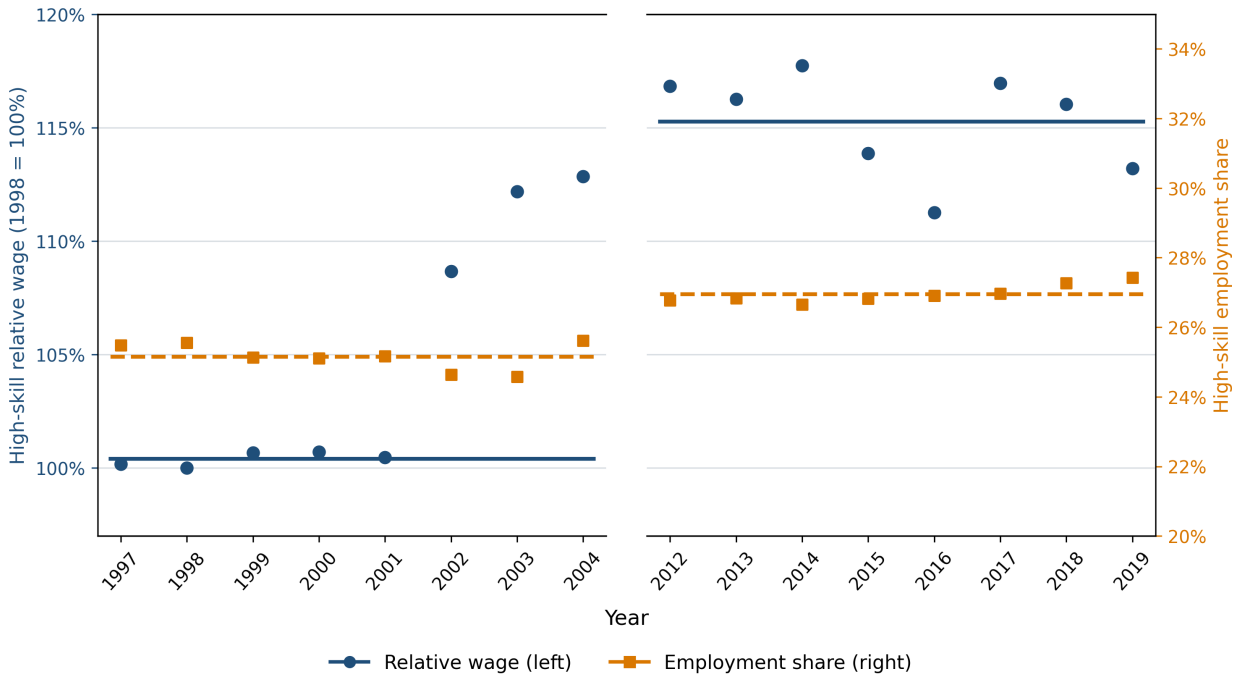
This pattern differs substantially from the evolution of the conventional college skill premium documented in the literature (see, for example, Autor et al. (2020)). The college skill premium increased persistently from the 1980s through the post-2010 period, even as the relative supply of college graduates also continued to rise. In contrast, my occupation-based classification identifies a smaller group of high-skill workers than the conventional group of college graduates and is associated with a larger relative wage differential.[4]

This definition differs from conventional education-based measures of skill primarily because it can capture changes in workers' skill requirements over the course of their careers. When skill is defined by educational attainment, an individual's skill category typically remains fixed unless the worker returns to school and obtains an additional degree. In contrast, the O*NET *Job Zone* classification is based more directly on the preparation, experience, and tasks required by a particular occupation. As workers accumulate experience, switch occupations, assume greater responsibilities, or move into more demanding roles, they may therefore move across skill categories even without acquiring additional formal education. This makes it important to measure skill at a detailed occupation-industry level and to allow for heterogeneity within broad occupational categories, so that changes in skill requirements over the worker's lifecycle can be reflected in the classification.

The O*NET *Job Zone* measure is therefore intended to identify occupations that are

[4]Acemoglu and Autor (2011) indicate the relative wage for college graduates is around 1.88 in 2003, while the relative wage in my calculation is around 2.1 in 2003.

Figure 1: Relative Wage Growth and Employment Share of High-Skill Workers



Notes: This figure plots the relative wage of high-skill workers (blue circles, left axis) and the high-skill employment share (orange squares, right axis). The relative wage is the average wage of high-skill workers relative to that of middle- and low-skill workers, normalized to 100% in 1998. The solid blue and dashed orange horizontal lines show the averages of the relative wage and the employment share, respectively, over 1997–2001 and 2012–2019. Skill groups are defined in Section 2.1. *Source:* OEWS and O*NET.

among the most skill-intensive jobs in the economy rather than to classify workers solely according to a fixed educational credential. I interpret high-skill workers in this framework as a relatively stable share of the workforce employed in occupations with the highest skill requirements. The following examples illustrate how this task- and occupation-based classification can differ from classifications based on education or broad occupational groups.

First, relative to education-based measures of skill, this occupation-based approach allows the classification to distinguish workers with similar educational backgrounds who perform jobs with very different skill requirements. Using 6-digit occupation codes, I differentiate occupations within broad job categories according to their specific tasks, responsibilities, and required preparation. For example, Legal Assistants (23-2011) and Title Searchers (23-2093), who provide administrative and support services to legal professionals, are classified as middle-skill workers under my Job Zone-based classification.

Although workers in these occupations often hold college degrees and work in settings associated with professional occupations, their job responsibilities more closely resemble administrative support tasks. The demand for these occupations has also followed patterns similar to those of other administrative and sales occupations, which are commonly classified as middle skill. Classifying these occupations as middle skill therefore more closely reflects the skill requirements of the jobs themselves rather than the educational attainment of the workers who hold them.

Conversely, the Job Zone classification places some occupations that would conventionally be grouped with middle- or lower-skill occupations into the high-skill category. For example, Insurance Sales Agents (41-3021) belong to the broader sales occupational category, which is often associated with middle-skill work. However, their tasks, including the use of financial analysis and medical software to customize insurance programs for individual customers, differ substantially from those performed in many other sales occupations, such as retail salespersons and travel agents.[5] O*NET therefore assigns Insurance Sales Agents to a higher skill category. Similarly, although most occupations within Transportation and Material Moving Occupations (53-0000) are classified as middle or low skill, Airline Pilots, Copilots, and Flight Engineers (53-2011) require substantially greater specialized training and professional skills and are therefore classified as high-skill labor.

2.2 Skill intensity in household consumption baskets

Skill intensity adjusted using the BEA input–output table. I divide workers into four skill groups: high-skill workers (H), middle-skill workers (M), low-skill service workers (L^S), and other low-skill workers (L^O). For each industry i and skill group h, I define the direct skill share, $\widetilde{\theta}_{ih}$, as the group's share of total labor compensation in 2003.[6] When multiple NAICS industries map to the same BEA input–output industry, I aggregate their skill shares using total labor compensation as weights. This procedure produces direct skill shares for 417 input–output industries.

The direct skill share captures only the composition of labor employed within an industry. To measure the skill intensity embodied in a final-demand category, I also account for the labor used to produce its intermediate inputs. I use the 2002 BEA commodity-by-

[5] Job descriptions are retrieved from O*NET: https://www.onetonline.org/link/summary/41-3021.00

[6] Wage bills are calculated using employment and annual mean wages from OEWS.

commodity input–output table and the official concordance between NAICS and BEA input–output industries to trace these indirect labor requirements throughout the production chain.[7]

Let ℓ_i denote industry i's direct labor cost share, and let Γ_{ij} denote the share of industry i's intermediate-input expenditure allocated to inputs produced by industry j. For each skill group h, the adjusted skill intensity of industry i, denoted by θ_{ih}, satisfies

$$\theta_{ih} = \ell_i \widetilde{\theta}_{ih} + (1 - \ell_i) \sum_{j=1}^{N} \Gamma_{ij} \theta_{jh}. \tag{1}$$

The first term captures industry i's direct use of skill group h, while the second captures the skill content embodied in its intermediate inputs.

In matrix form, equation (1) can be written as

$$\Theta_h = D(\ell)\widetilde{\Theta}_h + D(1 - \ell)\Gamma\Theta_h, \tag{2}$$

which implies

$$\Theta_h = \left[I - D(1 - \ell)\Gamma\right]^{-1} D(\ell)\widetilde{\Theta}_h. \tag{3}$$

Here, $\widetilde{\Theta}_h$ and Θ_h collect the direct and adjusted skill intensities, respectively, across the 417 industries; $D(\ell)$ is the diagonal matrix of labor cost shares; and I is the identity matrix. I apply this calculation separately to each of the four skill groups.

CEX consumption data. Finally, I map the adjusted skill intensity at the industry level to household-level consumption microdata from the Consumer Expenditure Survey (CEX). The CEX consists of two separate surveys. The Interview Survey collects major and recurring expenditures through quarterly interviews, while the Diary Survey records smaller and more frequently purchased items. I use the household-year expenditure data constructed by Hubmer (2023), which cover 1982 to 2016 and map more than 800 UCC consumption categories to the 2002 I-O table. To account for business cycle fluctuations and measurement error correlated with income, such as under-reporting of income by richer households and over-reporting by poorer households, I classify households into income

[7] Private households and government agencies reported in OEWS cannot be matched to NAICS industries and are therefore excluded. The final expenditures associated with these unmatched observations account for only a negligible share of aggregate expenditure.

percentiles using imputed permanent income, following Hubmer (2023).[8] I impute permanent income by regressing total expenditure on current after-tax income, education, and occupation.

For each household h in year t, the final dataset from CEX consists of the value and share of expenditure in each industry, and a vector of household characteristics that includes education, after-tax income and other demographic information. Given a household's expenditure share in each industry, I calculate the factor intensity of high, middle, low-skill service, and other low-skill workers in each household's consumption basket.

I group households into income percentile bins defined consistently across years, so that households in the same percentile bin have the same real income level regardless of the survey year. For each year, I compute the average skill intensity of the household consumption basket within each income bin. I then aggregate these values by decade, taking the average across years within each decade. Figure 2 plots the resulting decade-level profiles of skill intensity in household consumption baskets across the income distribution.

2.3 Consumption shifts toward skill-intensive goods and services as income rises

Using the final-good-level skill-intensity measures constructed above, I document novel patterns in the skill composition of household consumption across the income distribution and over time, shown in Figure 2. First, at any point in time, higher-income households consume a modest but systematically higher skill-intensity basket. In panel (a), moving from roughly the 10th to the 90th percentile of the income distribution raises the high-skill intensity of the consumption basket by about 1 percentage point. A similar, though somewhat smaller, pattern appears for low-skill service workers, as shown in panel (c). However, the opposite pattern holds for goods reliant on other types of low-skill workers (panel (d)): richer households spend less on goods and services that rely heavily on non-service low-skill workers. By contrast, the skill intensity of low-skill non-service workers in households near the 90th percentile is over 1 percentage point below that of households near the 10th percentile. Meanwhile, middle-skill intensity, shown

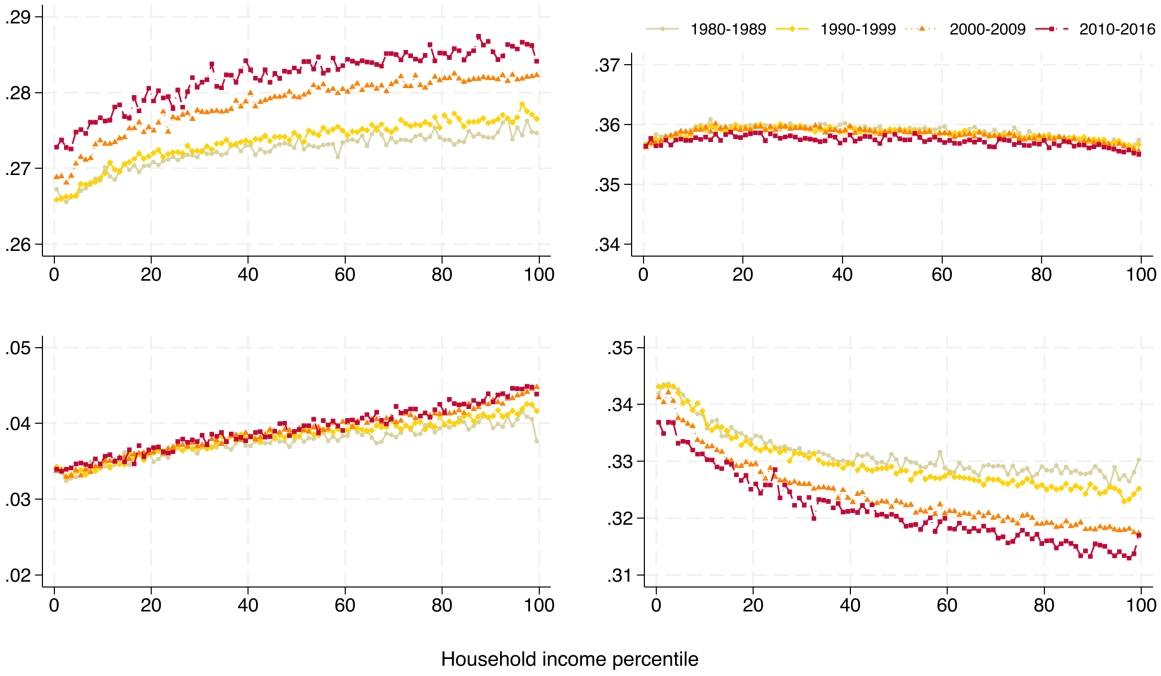
[8] For example, Carroll and Dunn (1997) find that consumption of durable goods is particularly sensitive to business cycle fluctuations. Since this paper focuses on the secular rise in wage inequality, I remove short-term business cycle fluctuations by using imputed permanent income.

in panel (b), is nearly flat across the income distribution. The magnitudes are quantitatively modest, in part because final-good skill intensity aggregates labor inputs across the entire production chain. Nevertheless, the income gradients are clear and remarkably systematic. They imply that, holding relative prices fixed, rising household income shifts expenditure toward goods and services that are relatively intensive in high-skill and low-skill service labor and away from those intensive in other low-skill labor, generating an income-driven change in relative labor demand.

The figure also reveals a distinct shift in the skill composition of consumption over time, even conditional on real income.[9] Across the income distribution, high-skill intensity in the household consumption basket rises by roughly 1 percentage point between the 1980s and 2010–2016, with much of the increase occurring from the 1990s onward. The corresponding movement for consumption in non-service low-skill workers (panel (d)) goes in the opposite direction: the intensity of non-service low-skill labor falls by roughly 1 percentage point or more over the same period, with a larger decline among the right tail of the income distribution. In comparison, changes in middle-skill and low-skill service intensity over time are small. Note that since the skill intensity is fixed at the 2003 level and the household income is fixed at the same level for each point on the x-axis, the shifts over time capture only the consumption shifts across industries. While the magnitudes are not large, the reallocation over time is visible: conditional on income, household expenditure has gradually shifted toward goods and services that are more intensive in high-skill labor and away from those intensive in non-service low-skill labor. This pattern likely reflects technological change that shifts the relative prices of goods, thus shifting their relative consumption shares through a price effect.

[9] The bins correspond to fixed real-income levels across survey years.

Figure 2: Household Income Percentiles and Skill Intensity in the Consumption Basket



Notes: This figure shows the skill intensity embodied in household consumption baskets across the income distribution over time, where skill intensity is measured as the wage-bill share of each skill group using O*NET linked with OEWS data. The classification of skill levels is described in detail in the Empirics section. To account for measurement error in individual goods and the presence of durable goods, income is imputed using education, after-tax income, and occupation, following Hubmer (2023). Income percentiles are defined to be stable over time, so that households in a given income-percentile bin have the same real income in any survey year. *Source:* O*NET, OEWS, BEA I–O Tables, and CEX.

2.4 Expenditure elasticities and skill intensities

Figure 3 plots average expenditure elasticities against high-skill factor intensities across final-demand industries. Good-level expenditure elasticities and labor shares are taken from the baseline estimates of Hubmer (2023). High-skill factor intensity is defined as the labor share of total production costs multiplied by the high-skill share of the wage bill, with both components accounting for direct and indirect inputs throughout the production chain. The measure therefore captures both the skill composition of labor and the overall labor intensity of production: a low value may reflect either greater reliance on lower-skill labor or greater capital intensity. Marker size indicates each category's share of consumption expenditure.

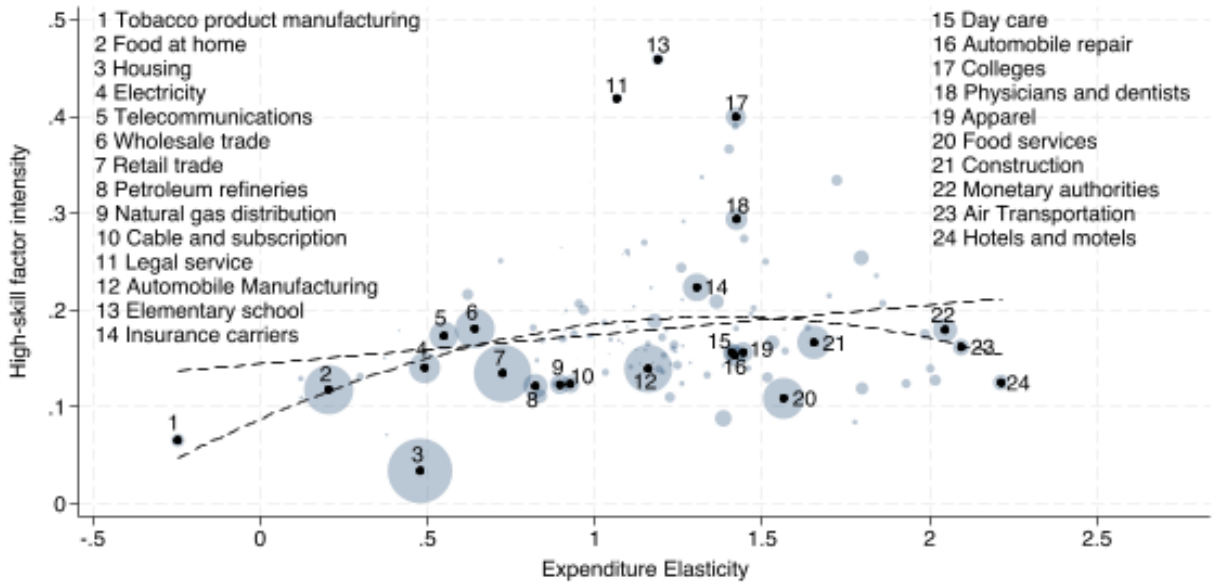
Overall, expenditure elasticity and skill intensity are positively related, although there is substantial heterogeneity across industries. Industries with relatively low expenditure elasticities, such as tobacco products, food at home, and housing, tend to have low skill intensity, whereas several industries with expenditure elasticities above one, including legal services, educational institutions, physicians and dentists are among the most skill intensive. The relationship is not monotonic, however: some industries with the highest expenditure elasticities, including travel-related categories such as hotels and air transportation, have only moderate or low skill intensity.

The overall pattern suggests that, as income rises and consumption shifts toward goods and services with higher expenditure elasticities, expenditure tends to be reallocated toward more skill-intensive sectors. This is consistent with the evidence presented in the top-left panel of Figure 2. This relationship between expenditure patterns and skill intensity motivates the introduction of nonhomothetic preferences in the model developed in the next section.

3 Model

Motivated by the empirical patterns documented above, I develop a general equilibrium model with nonhomothetic preferences to capture shifts in the composition of consumption. The model also features capital-skill complementarity and industry-specific productivity growth to capture differential technological change across industries. Within this framework, the evolution of the relative wage is driven by four forces: aggregate productivity growth, industry-specific productivity growth, the accumulation of equipment

Figure 3: Expenditure Elasticities and High-Skill Factor Intensity



Notes: This figure plots industry-level expenditure elasticities against high-skill factor intensity for the final-demand categories in the CEX. High-skill factor intensity is calculated as the labor share in total production costs multiplied by the high-skill share of the wage bill, accounting for labor and capital used both directly and indirectly throughout the production chain. Marker size represents the category's share of consumption expenditure. *Source:* O*NET, OEWS, BEA input-output table, CEX.

capital, and the accumulation of structures capital.

3.1 Production

In the market, there are I industries, each producing a final good under perfect competition. I assume that the production process for each good follows a Cobb-Douglas production function with respect to structures capital (S), and a nested CES production function for the other inputs, including skilled workers (H), low-skill workers (L), and capital equipment (K).

There are two ways to nest K, H, and L within a CES production function that allow for capital-skill complementarity. The first method combines K and L before combining their aggregate with H, restricting the elasticity of substitution between H and L to be the same as that between H and K. The second method combines K and H before combining their aggregate with L, restricting the elasticity of substitution between L and H to be the same as that between L and K. In both specifications, capital equipment and low-skill

labor are assumed to be closer substitutes compared to capital equipment and skilled labor. The difference lies in the elasticity of substitution between labor types. In the first specification, skilled and low-skill labor complement each other, whereas in the second specification they are substitutes. Skilled labor in my data focuses on more abstract tasks and assumes more managerial responsibilities that are distinct from the routine tasks performed by low-skill labor. Therefore, I prefer the first modeling functional form where high- and low-skill workers are complements.[10]

The production function of industry i is given by:

$$Y_{it} = A_{it} \left(S_{it}\right)^{\beta_i} X_{it}^{1-\beta_i} \tag{4}$$

where,

$$X_{it} = \left[{\alpha_i}^{\frac{1}{\eta}} H_{it}^{\frac{\eta-1}{\eta}} + (1-\alpha_i)^{\frac{1}{\eta}} \left({\delta_i}^{\frac{1}{\rho}} L_{it}^{\frac{\rho-1}{\rho}} + (1-\delta_i)^{\frac{1}{\rho}} \left(K_{it}\right)^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \tag{5}$$

A_{it} is factor-neutral industry-specific productivity. β_i, δ_i and α_i are industry-specific parameters that govern factor shares, assumed to be constant over time. η is the elasticity of substitution between high-skill labor and the low-skill–equipment composite, while ρ is the elasticity of substitution between equipment and low-skill labor. Capital-skill complementarity requires $\eta < \rho$. When η and ρ approach 1, we have the standard Cobb-Douglas function. All production factors are measured in efficiency units.

Firm's problem. The representative firm in industry i takes the wages W_t^H, W_t^L and the rental rates of capital R_t^E, R_t^S as given, and maximizes its profit:

$$\underset{H_{it},\, L_{it},\, K_{it},\, S_{it}}{Max} P_{it}Y_{it} - W_t^H H_{it} - W_t^L L_{it} - R_t^E K_{it} - R_t^S S_{it} \tag{6}$$

Then from the firm's optimization condition, the final good price in industry i can be

[10] See Krusell et al. (2000) and Parro (2013) for discussion on the second specification.

written as

$$P_{it} = \frac{1}{A_{it}} \left(\frac{R_t^S}{\beta_i} \right)^{\beta_i} \left(\frac{P_{it}^X}{1-\beta_i} \right)^{1-\beta_i} \tag{7}$$

where P_{it}^X denotes the factor price of X_{it}, and can be written as

$$P_{it}^X = \left[\alpha_i \left(W_t^H \right)^{1-\eta} + (1-\alpha_i) \left[\delta_i \left(W_t^L \right)^{1-\rho} + (1-\delta_i) \left(R_t^E \right)^{1-\rho} \right]^{\frac{1-\eta}{1-\rho}} \right]^{\frac{1}{1-\eta}} \tag{8}$$

There are three production-side channels through which technological change affects relative industry prices and, consequently, the composition of expenditure across industries. First, differential industry-level productivity growth lowers the relative price of goods produced in industries experiencing faster productivity growth. Second, changes in factor prices have heterogeneous effects on production costs depending on industries' factor intensities. For example, holding wages fixed, a decline in the rental price of equipment capital lowers production costs more in equipment-intensive industries and therefore reduces their relative output prices. Third, equipment accumulation affects relative wages through capital–skill complementarity. As equipment becomes more abundant and its rental price falls, firms substitute equipment for low-skill labor while increasing their demand for high-skill labor. This raises the relative wage of high-skill workers and, in turn, increases the relative price of goods produced in skill-intensive industries.

3.2 Demand

Each worker is endowed with capital $\bar{S}$ and $\bar{K}$ and one unit of labor. The return on capital is assumed to be equally distributed across all households.

A fraction f of the population is high skill, while the remaining $1-f$ is low skill. I assume a log demand function that can be implied by some underlying primitive utility function.[11] All consumers have the same preferences and therefore face the same compensated elasticity of substitution, σ, and the same good-specific expenditure elasticity,

[11] Log-demand offers the advantage of flexibility in modeling income effects. Examples using log-demand include Goos et al. (2014) for homothetic preferences and Hubmer (2023) for nonhomothetic preferences.

γ_{it}. At any point in time, differences in their expenditure shares arise solely from differences in income. The expenditure share of good i for household h is $\omega_{it}^h = \frac{C_{it}^h P_{it}}{E_t^h}$, where this share is exogenously given in the initial period and evolves over time according to

$$d\ln\omega_{it}^h = (1-\sigma)d\ln\frac{P_{it}}{P_t^h} + (\gamma_{it}-1)d\ln\frac{E_t^h}{P_t^h}, \quad h \in \{H, L\} \tag{9}$$

where the change in household-level income and aggregate price index are given by:

$$E_t^h = W_t^h + R_t^E\bar{K}_t + R_t^S\bar{S}_t \tag{10}$$

$$d\ln P_t^h = \sum_i \omega_{it}^h d\ln P_{it} \tag{11}$$

Equation 9 illustrates that the reallocation in consumption across industries is driven by changes in the relative prices of goods and changes in total expenditure. In a model with homothetic preferences, the expenditure elasticity γ_{it} equals one for all industries at all times. Thus, the substitution effect driven by changes in relative prices from the production side is the only source of industry reallocation. When goods are gross substitutes, as in the estimated demand system ($\sigma > 1$), expenditure shifts toward industries experiencing larger relative price declines. In a model with nonhomothetic preferences, however, households also shift their consumption towards industries with higher expenditure elasticities as total expenditures increase. Industries with $\gamma_{it} > 1$ are luxury goods whose expenditure shares increase with total expenditure, while industries with $\gamma_{it} < 1$ are necessity goods whose expenditure shares decrease with total expenditure.[12]

Thus, technological change affects the allocation of expenditure across industries through two margins: a substitution effect arising from changes in relative goods prices and, under nonhomothetic preferences, an income effect arising from growth in real expenditure.

3.3 Equilibrium

I use the wage of low-skill workers as the numeraire, normalizing $W_t^L = 1$. Solving for equilibrium requires us to find the rental rates of capital (R_t^E, R_t^S) and the wage premium W_t^H such that, given factor prices and the implied goods prices, all factor and goods

[12] I follow Hubmer (2023) and use time-varying industry-level expenditure elasticities, but estimation shows that they are very stable over time. See Section 5.3 in Hubmer (2023) for more details.

markets clear.

More specifically, the competitive equilibrium consists of factor prices $\{R_t^S, R_t^E, W_t^H\}$, goods prices $\{P_{it}\}_{i\in I}$, consumer demand for both labor types $\{C_{it}^H, C_{it}^L\}_{i\in I}$, total expenditures $\{E_t^H, E_t^L\}$, final-good outputs $\{Y_{it}\}_{i\in I}$, and factor input choices $\{H_{it}, L_{it}, K_{it}, S_{it}\}_{i\in I}$, such that:

1. Consumer demand is given by $C_{it}^h = \frac{\omega_{it}^h E_t^h}{P_{it}}$, for $h \in \{H, L\}$, where ω_{it}^h is exogenously given at $t = 0$ and evolves according to equation 9;

2. Given goods prices $\{P_{it}\}_{i\in I}$ and factor prices W_t^H, R_t^E, and R_t^S, final-good outputs $\{Y_{it}\}_{i\in I}$ and factor input choices $\{H_{it}, L_{it}, K_{it}, S_{it}\}_{i\in I}$ are consistent with profit maximization subject to equation 4;

3. All final-goods markets clear: $fC_{it}^H + (1-f)C_{it}^L = Y_{it}$ for all $i \in I$;

4. All factor markets clear: $f = \sum_{i\in I} H_{it}$, $1-f = \sum_{i\in I} L_{it}$, $\bar{K}_t = \sum_{i\in I} K_{it}$, and $\bar{S}_t = \sum_{i\in I} S_{it}$.

3.4 Mechanisms

Four exogenous forces shape the evolution of the skill premium in the model: aggregate productivity growth, the accumulation of structures capital, industry-specific productivity growth, and the accumulation of equipment capital. These forces affect the relative demand for high- and low-skill labor through a combination of income effects, relative-price changes, and capital-skill complementarity.

First, aggregate productivity growth, A_t, affects the skill premium primarily through the income channel generated by nonhomothetic preferences. An increase in aggregate productivity raises real household expenditure without directly changing relative productivity across industries. As real expenditure rises, households reallocate expenditure toward goods and services with higher expenditure elasticities. When industries with higher expenditure elasticities are also more skill intensive, this reallocation increases the relative demand for high-skill labor and raises the skill premium. Figure 3 documents the positive relationship between industry-level expenditure elasticities and skill intensity that underlies this mechanism.

Second, the accumulation of structures capital, S_t, affects the skill premium through both an income channel and a relative-price channel. An increase in structures capital raises aggregate productive capacity and household income, thus generating a non-

homothetic reallocation of expenditure similar to that induced by aggregate productivity growth. At the same time, because industries differ in their structures intensity, β_i, changes in the rental price of structures affect production costs differently across industries. The resulting changes in relative goods prices induce further reallocation of expenditure across industries and therefore alter relative labor demand according to the skill intensity of the industries that expand.

Third, industry-specific productivity growth, A_{it}, affects the skill premium through a substitution effect and changes in relative goods prices and the resulting reallocation of expenditure across industries. Industries experiencing faster productivity growth have larger declines in their relative output prices. Since final goods are gross substitutes across industries, expenditure shifts toward industries experiencing larger productivity-driven price declines. If these industries are also more skill intensive, this reallocation increases the relative demand for high-skill labor and raises the skill premium. Furthermore, if these industries are more capital-intensive, this reallocation reinforces the capital-skill complementarity channel and increases the skill premium. Figure 5a documents the relationship between industry-specific productivity growth and skill intensity.

Finally, the accumulation of equipment capital, K_t, affects the skill premium mainly through capital-skill complementarity. As equipment becomes cheaper, firms substitute equipment for low-skill labor while increasing their demand for high-skill labor, which is relatively complementary to equipment. This shifts relative labor demand toward high-skill workers and raises their relative wage. Equipment accumulation also changes relative goods prices because the decline in the rental price of equipment lowers production costs more strongly in equipment-intensive industries. Moreover, this channel can interact with industry-specific productivity growth: if faster-growing industries are also more equipment intensive, their expansion further increases the demand for equipment and, in turn, for high-skill labor because of capital-skill complementarity. Figure 5b documents the positive relationship between industry-level productivity growth and equipment intensity.

4 Estimation and Calibration

I calibrate the model to the period from 1982 to 2019 to best capture the increase in the stock of capital since the 1980s. To quantify the impact of various channels within the model, I first estimate or calibrate the model parameters. These include the preference

parameters $\{\gamma_{it}, \sigma\}_{i \in I,\, t \in [1982,2019]}$, technology parameters $\{\beta_i, \alpha_i, \delta_i, \eta, \rho, A_{it}\}$, and consumption shares at the initial steady state for both types of labor $\{\omega_{it}^H, \omega_{it}^L\}_{i \in I,\, t=1982}$, the stocks of capital structures and capital equipment $\{S_t, K_t\}_{t \in [1982,2019]}$, and the fraction of high-skill workers in the economy f.

The sources and summary statistics for these parameters are reported in Table 1. I begin by estimating the two substitution elasticities governing capital–skill complementarity, followed by a description of the calibration of the remaining parameters. I use the demand-side elasticities estimated by Hubmer (2023). Finally, I isolate each channel and study their individual contributions to the rise in the skill premium.

Table 1: Calibration

Parameter	Moment/Description	Value	Source
ρ	Elas of substitution between K and L	1.234	My estimation
η	Elas of substitution between H and the L–K composite	0.414	My estimation
δ_i	Governs the L share in production in 2003	0.782[a]	BEA I–O, OEWS, O*NET
α_i	Governs the H share in production in 2003	0.162[a]	BEA I–O, OEWS, O*NET
β_i	Governs the S share in production in 2003	0.173[a]	BEA I–O, OEWS, O*NET
f	Share of H workers in 2003	0.25	OEWS, O*NET
σ	Elasticity of substitution between goods	1.55	Hubmer (2023)
γ_{it}	Expenditure elasticity for Y_i		Hubmer (2023)
$\omega_{i,1982}^H$	Initial consumption share of Y_i for H workers[b]		CEX
$\omega_{i,1982}^L$	Initial consumption share of Y_i for L workers		CEX

[a] Reported values are time averages of the consumption-expenditure-weighted mean.
[b] Calculated using consumption of households in the top 30% of the income distribution.

4.1 Estimation of capital-skill complementarity

The estimation exploits heterogeneous industry exposure to long-run changes in factor prices. Industries differ in their factor intensities, so the secular decline in the relative price of equipment generates differential changes in production costs across industries. I use this variation to identify the relative substitutability of equipment with high- and low-skill labor. The empirical strategy builds on Karabarbounis and Neiman (2014), Raval (2019), and Hubmer (2023), who use changes in factor prices and factor shares to estimate elasticities of substitution.

The estimation exploits secular movements in factor prices rather than cross-sectional variations, allowing the estimated elasticities to incorporate longer-run adjustments in

the production and the reallocation of tasks across skill groups and between labor and equipment. I estimate the elasticities using annual data from 2002 to 2019, and I use FRED series PERIC to measure changes in the relative price of equipment.[13] I construct industry exposure to changes in factor prices using the factor shares, while identification comes from the interaction of these heterogeneous exposures with changes in factor prices over time. The resulting estimates should therefore be interpreted as long-run substitution elasticities and may differ from estimates identified from short-run variation.

The regression is constructed at the level of final-demand industries, with factor shares accounting for inputs used both directly and indirectly through intermediate goods along the production chain. This corresponds to the *final expenditure approach* in Herrendorf et al. (2013), in contrast to the *value-added approach*, which assigns factor use to the industries in which value added is directly produced. I therefore treat each final-demand industry as a vertically integrated production sector, with its factor intensities reflecting the full production chain. The estimated elasticities capture adjustments both within final-good industries and through changes in the composition of intermediate inputs.

To allow for unobserved factor-augmenting technological change, I express labor and equipment inputs in efficiency units. Let

$$H_{it} = A_{it}^{H} h_{it}, \qquad L_{it} = A_{it}^{L} l_{it}, \qquad K_{it} = A_{it}^{E} k_{it}^{E}, \qquad S_{it} = k_{it}^{S},$$

where A_{it}^{H}, A_{it}^{L}, and A_{it}^{E} denote factor-augmenting technologies, and h_{it}, l_{it}, k_{it}^{E} and k_{it}^{S} denote observed factor inputs. The corresponding effective factor prices are

$$\frac{W_t^H}{A_{it}^H}, \qquad \frac{W_t^L}{A_{it}^L}, \qquad \frac{R_t^E}{A_{it}^E}.$$

Factor-augmenting technological change is not separately observed and enters the residuals of the estimating equations below. The first-order conditions from the firm's profit-maximization problem imply the following relationships for factor shares:

$$\theta_{it}^{S} = \frac{R_t^S k_{it}^S}{P_{it} Y_{it}} = \beta_i \tag{12}$$

[13] This sample period is constrained by the OEWS data. The first year using the NAICS industry classification is 2002. The BLS explicitly identifies this transition as a break affecting the historical comparability of industry-level estimates, so I begin the estimation in 2002.

$$\theta_{it}^{H} = \frac{W_t^H h_{it}}{P_{it} Y_{it}} = \frac{\alpha_i}{1-\alpha_i} \left(\frac{P_{it}^M}{W_t^H / A_{it}^H} \right)^{\eta-1} \theta_{it}^{M} \tag{13}$$

$$\theta_{it}^{L} = \frac{W_t^L l_{it}}{P_{it} Y_{it}} = \delta_i \left(\frac{P_{it}^M}{W_t^L / A_{it}^L} \right)^{\rho-1} \theta_{it}^{M} \tag{14}$$

$$\theta_{it}^{E} = \frac{R_t^E k_{it}^E}{P_{it} Y_{it}} = (1-\delta_i) \left(\frac{P_{it}^M}{R_t^E / A_{it}^E} \right)^{\rho-1} \theta_{it}^{M} \tag{15}$$

where $\theta_{it}^{M} = \frac{P_{it}^M M_{it}}{P_{it} Y_{it}} = (1-\alpha_i)(1-\beta_i) \left(\frac{P_{it}^X}{P_{it}^M} \right)^{\eta-1}$. By construction, $\theta_{it}^{S} + \theta_{it}^{E} + \theta_{it}^{H} + \theta_{it}^{L} = 1$ and $\theta_{it}^{E} + \theta_{it}^{L} = \theta_{it}^{M}$.

4.1.1 Estimation of $\rho - \eta$

I first estimate the difference between ρ, which governs substitution between low-skill labor and equipment, and η, which governs substitution between high-skill labor and the low-skill–equipment composite. Taking the log ratio of equations 14 and 13 gives

$$\ln\left(\frac{\theta_{it}^{L}}{\theta_{it}^{H}}\right) = \ln\left(\frac{(1-\alpha_i)\delta_i}{\alpha_i}\right) + (\rho-\eta)\ln P_{it}^{M} + (\eta-1)\ln\left(\frac{W_t^H}{A_{it}^H}\right) + (1-\rho)\ln\left(\frac{W_t^L}{A_{it}^L}\right). \tag{16}$$

Equation 16 links relative factor shares to the cost of the low-skill–equipment composite. When $\rho > \eta$, a decline in P_{it}^M lowers $\theta_{it}^L / \theta_{it}^H$, increasing the factor share of high-skill labor relative to low-skill labor. In the data, changes in P_{it}^M are driven primarily by the secular decline in the relative price of equipment. The log change in the price of the composite can be written as

$$\hat{p}_{it}^{M} = \frac{\theta_i^L}{\theta_i^M}\left(\hat{w}_t^L - \hat{a}_{it}^L\right) + \frac{\theta_i^E}{\theta_i^M}\left(\hat{r}_t^E - \hat{a}_{it}^E\right), \tag{17}$$

where $\hat{Z} = \ln Z_{t'} - \ln Z_t$ denotes the log change in variable Z between periods t and t'. Substituting equation 17 into equation 16, and absorbing common wage movements and time-invariant industry characteristics with fixed effects, yields the estimating equation

$$\ln\left(\frac{\theta_{it}^L}{\theta_{it}^H}\right) = \widetilde{\alpha}_i + \lambda_t + (\rho - \eta)\left[\frac{\theta_i^L}{\theta_i^M}\hat{w}_t^L + \frac{\theta_i^E}{\theta_i^M}\hat{r}_t^E\right] + \xi_{it}, \tag{18}$$

where $\widetilde{\alpha}_i$ denotes industry fixed effects, λ_t denotes time fixed effects, and ξ_{it} contains industry-specific factor-augmenting technological change and other unobserved shocks.

Identification comes from heterogeneous industry exposure to common movements in factor prices. The estimating equation has a shift-share structure: aggregate changes in low-skill wages and equipment rental prices provide the common shifts across industries, while initial industry factor intensities determine each industry's exposure to these changes. In the baseline specification, I measure these exposures using factor shares in 2002, prior to the subsequent secular decline in the relative price of equipment. For example, a decline in the rental price of equipment generates a larger predicted decline in P_{it}^M for industries with a higher initial equipment share, $\theta_{i,2002}^E/\theta_{i,2002}^M$. Aggregate shocks and the common movements in factor prices across industries are absorbed by the time fixed effects. $\rho - \eta$ is identified from differential changes in relative factor shares across industries with different initial exposures.

The key exclusion restriction is that, conditional on industry and time fixed effects, initial factor intensities are not systematically correlated with subsequent industry-specific factor-augmenting technological change contained in ξ_{it}. In particular, industries that were initially more exposed to the decline in equipment prices should not subsequently experience systematically different unobserved skill-, low-skill-, or equipment-augmenting technological change. Under this assumption, the interaction between common factor-price changes and initial industry exposure identifies $\rho - \eta$.

A potential threat to this identification strategy is industry-specific factor-augmenting technological change that is systematically related to initial factor intensities. For example, if initially equipment-intensive industries subsequently experience faster equipment-

augmenting technological change, their effective equipment prices would decline by more than implied by the observed aggregate equipment-price series. The estimating equation would then attribute part of the resulting change in relative factor shares to industries' differential exposure to the observed decline in equipment prices. Because factor-augmenting technologies are not directly observed, I cannot directly test this exclusion restriction.[14] For comparison, I also construct exposure measures using factor shares averaged over the sample period.

4.1.2 Estimation of $\eta - 1$

I next estimate $\eta - 1$, which governs the elasticity of substitution between high-skill labor and the low-skill–equipment composite. Unlike the estimation of $\rho - \eta$, which uses the relative factor shares of low- and high-skill labor, this estimation is based only on the factor share of high-skill labor. Substituting θ_{it}^M into equation 13 and taking logs gives

$$\ln\left(\theta_{it}^H\right) = \ln\left[\alpha_i(1-\beta_i)\right] + (\eta-1)\ln P_{it}^X - (\eta-1)\ln\left(\frac{W_t^H}{A_{it}^H}\right). \tag{19}$$

The equation above links the high-skill factor share to the cost of the composite input X_{it}. When $\eta < 1$, high-skill labor is relatively complementary to the low-skill–equipment composite. A decline in P_{it}^X therefore raises the factor share of high-skill labor, holding the effective high-skill wage fixed. The log change in the price of X_{it} can be written as

$$\begin{aligned}\hat{p}_{it}^X &= \frac{\theta_i^H}{\theta_i^M + \theta_i^H}\left(\hat{w}_t^H - \hat{a}_{it}^H\right) + \frac{\theta_i^M}{\theta_i^M + \theta_i^H}\hat{p}_{it}^M \\ &= \frac{\theta_i^H}{1-\beta_i}\left(\hat{w}_t^H - \hat{a}_{it}^H\right) + \frac{\theta_i^L}{1-\beta_i}\left(\hat{w}_t^L - \hat{a}_{it}^L\right) + \frac{\theta_i^E}{1-\beta_i}\left(\hat{r}_t^E - \hat{a}_{it}^E\right).\end{aligned} \tag{20}$$

Substituting this expression into the high-skill factor-share equation yields the estimating equation:

[14] The positive correlations between factor intensities and industry-specific TFP growth shown in Figure 5a do not mechanically violate this restriction. The calibrated A_{it} shown in Figure 5a captures factor-neutral productivity growth rather than the unobserved factor-augmenting technologies contained in ξ_{it}. In particular, this measure is constructed from changes in relative industry prices after accounting for changes in input prices.

$$\ln\left(\theta_{it}^{H}\right) = \widetilde{\alpha}_i + \lambda_t + (\eta - 1)\left[\frac{\theta_i^H}{1-\beta_i}\hat{w}_t^H + \frac{\theta_i^L}{1-\beta_i}\hat{w}_t^L + \frac{\theta_i^E}{1-\beta_i}\hat{r}_t^E\right] + \xi_{it}, \tag{21}$$

where $\widetilde{\alpha}_i$ denotes industry fixed effects, λ_t denotes time fixed effects, and ξ_{it} contains industry-specific factor-augmenting technological change and other unobserved shocks.

The identification strategy follows the same shift-share logic and relies on the same exclusion restriction as in the estimation of $\rho - \eta$, using initial 2002 factor shares as exposure measures. I also report estimates using sample-average factor shares for comparison.

Table 2 reports the estimates. Using initial 2002 factor shares, the estimated $\rho - \eta$ is 0.8197, with a standard error of 0.3412, and the estimate of $\eta - 1$ is -0.5856, with a standard error of 0.2675. Both are statistically significant at the five percent level. These estimates imply that $\eta = 0.4144$ and $\rho = 1.2341$. Thus, the baseline estimates imply $\rho > \eta$ and $\rho > 1 > \eta$, consistent with capital-skill complementarity: equipment is relatively substitutable with low-skill labor but complementary to high-skill labor. Using sample-average factor shares produces similar estimates, with $\rho - \eta = 0.782$, $\eta = 0.476$, and $\rho = 1.258$.

Table 2: Estimates of Capital–Skill Complementarity

	Initial 2002 Shares (1)	**Sample-Average Shares** (2)
$\rho - \eta$	0.8197**	0.7823**
	(0.3412)	(0.3284)
$\eta - 1$	-0.5856**	-0.5243*
	(0.2675)	(0.2826)
N	2128	2128

Notes: This table reports estimates of $\rho - \eta$ and $\eta - 1$ using annual data from 2002 to 2019. Column (1), the baseline specification, uses initial 2002 factor shares to construct industry exposure measures. Column (2) uses factor shares averaged over the sample period. All specifications include time and industry fixed effects and weight goods by their final-demand expenditure shares. Standard errors, reported in parentheses, are clustered at the good level. High-skill wage-bill shares and wages are constructed from OEWS-O*NET data as described in Section 2. Labor shares and equipment intensities are taken from the baseline estimates in Hubmer (2023). The high-skill factor share used to estimate $\eta - 1$ is constructed as the high-skill wage-bill share multiplied by the industry-level labor share. Changes in the rental price of equipment are proxied using the relative price of equipment from FRED series PERIC.

4.1.3 Comparison with the prior literature

Despite a large literature estimating the elasticity of substitution across different types of labor (e.g., Card (2009), Ottaviano and Peri (2012), Raveh and Reshef (2016)) and between capital and labor (e.g., Chirinko (2008), Karabarbounis and Neiman (2014), Oberfield and Raval (2021)), estimates of the elasticity of substitution between capital and labor by skill level remain relatively scarce. A key empirical challenge is that such estimation requires information on changes in factor shares together with changes in capital returns and wages by skill group for a common set of industries.

An important exception is Krusell et al. (2000), who estimate these substitution elasticities jointly with factor-augmenting technological change and factor-share parameters using a simulated pseudo-maximum likelihood estimator applied to annual U.S. data from 1963 to 1992. They estimate an inner-nest elasticity of 0.67 between skilled labor and equipment capital and an outer-nest elasticity of 1.67 between low-skill labor and the skilled-labor–equipment composite. Ohanian et al. (2023) re-estimate the same framework using data extended through 2019 and obtain corresponding elasticities of 0.75 and 1.70. My estimates imply lower substitution elasticities than those in Krusell et al. (2000), while preserving the pattern that equipment is more substitutable with low-skill labor than with high-skill labor. My empirical approach differs from these studies along two main dimensions. First, I define skill using O*NET occupational requirements, which isolates a relatively stable share of workers employed in the most skill-intensive occupations rather than relying on educational attainment. Second, I conduct the estimation at the level of final-demand industries and measure factor intensities using the full factor content of production, including intermediate inputs along the production chain. This corresponds to the final-expenditure approach described above and differs from approaches that assign factors only to the industries in which value added is directly produced. In the quantitative exercises, I also repeat the decomposition using the elasticities estimated by Krusell et al. (2000) as a robustness check in the Appendix.

My empirical strategy is also closely related to Karabarbounis and Neiman (2014), who estimate long-run substitution between capital and labor by exploiting variation in the relative price of investment goods. Their baseline identification relies on cross-country variation in long-run changes in investment prices and labor shares, and they also conduct industry-level exercises using 10 broad KLEMS sectors across 14 developed economies. They extend the two-factor capital-labor specification to a nested CES with high-skill and low-skill workers, allowing changes in skill composition to affect the rela-

tionship between investment prices and the aggregate labor share. However, their empirical specification identifies only the outer-nest elasticity; the inner elasticity governing substitution between capital and the skill type with which it is nested is not separately identified. In contrast, my approach separately identifies $\rho - \eta$ and $\eta - 1$, allowing me to recover both substitution elasticities. Across their nested-CES specifications, the estimated outer-nest elasticity ranges from 1.19 to 1.34, with an average of 1.26. While this parameter is not directly comparable to my estimate of ρ, the magnitude is close to my estimate of the elasticity between low-skill labor and equipment capital, $\rho = 1.258$, obtained using sample-average factor shares.

4.2 Remaining parameters

The expenditure elasticities γ_{it} and price elasticity σ are adopted directly from the estimation conducted by Hubmer (2023) using cross-sectional CEX data. I allow good-level expenditure elasticities to vary over time, although the variation is small. The initial expenditure share for industry i for high-skill workers, denoted as $\omega^H_{i,1982}$, is computed using the mean of the top 30% of households in the CEX, while $\omega^L_{i,1982}$ is defined as the mean expenditure share of the remaining households. The fraction of high-skill workers, denoted as f, is determined by the fraction of high-skill workers employed in 2003 calculated using OEWS-O*NET data.

I calibrate the industry-level production parameters β_i, α_i and δ_i to match the factor shares in 2003. The model assumes β_i, α_i and δ_i to be constant over time, and allows factor shares to adjust in response to the exogenous evolution of equipment capital through capital–skill complementarity.

The remaining model parameters are the factor-neutral productivity series A_{it}, and the evolution in capital stock for equipment and structures K_t, S_t. The stock of capital evolves according to the data. Capital equipment K includes equipment and machinery as well as intangible assets such as software (FRED series: K1NTOTL1EQ000). Structures capital S accumulates the cost of construction activity such as overhead and office costs, as well as the cost of materials and engineering, but does not include the cost of residential buildings (FRED series: K1NTOTL1ST000) (Lally (2009)). Note that in the model, an increase in the capital stock is isomorphic to increasing the capital-augmenting technology.

The residual component of output growth is attributed to total factor productivity. I first calibrate a common productivity component, A_t, that is shared across industries and

chosen to match the evolution of real per capita GDP (FRED series: A939RX0Q048SBEA).[15] I then allow productivity growth to vary across industries. Relative changes in industry prices are used to discipline differences in productivity growth across industries, while the aggregate component is jointly chosen so that the model continues to match real per capita GDP.[16] The appendix reports the calibrated productivity growth for all 133 industries. The fastest-growing industries are concentrated in technology-intensive manufacturing sectors, including Measuring and Controlling Device and Electronic Computer Manufacturing.

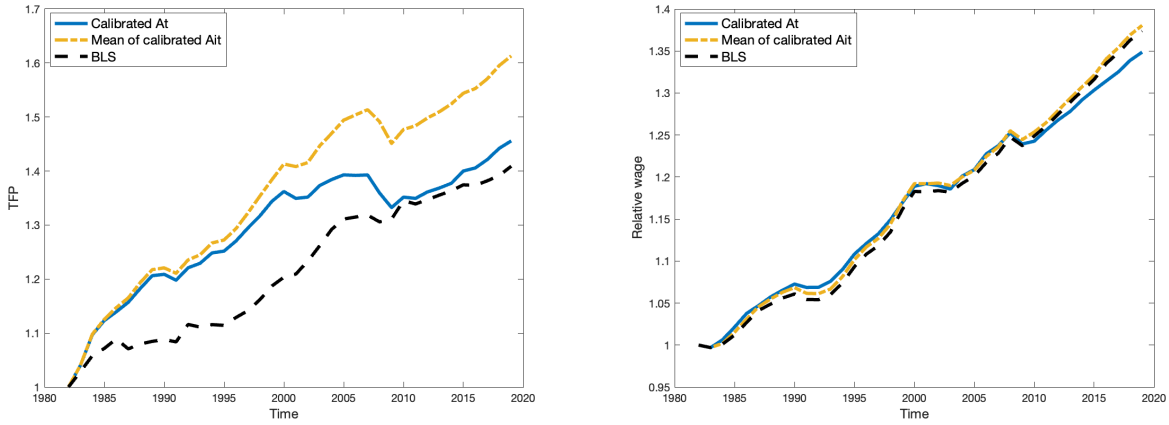
Figure 4a compares the resulting model-implied productivity series with the TFP series reported by the BLS (FRED series: MFPPBS). The solid line shows the calibrated common productivity component A_t, while the dashed line reports the value-added-weighted average of calibrated industry-specific productivity, $A^{GDP}it$. Both calibrations match the growth of real per capita GDP, but the latter additionally matches changes in relative industry prices. The BLS series provides an external benchmark for the aggregate evolution of productivity. In the quantitative analysis, I therefore consider three alternative productivity measures: the calibrated common component A_t, the industry-specific series $A^{GDP}it$ calibrated to match both relative prices and real per capita GDP, and an alternative industry-specific series A_{it}^{BLS} that matches relative price changes while aligning aggregate productivity growth with the BLS series.

Figure 4b compares the model-implied evolution of the relative wage of high-skill workers under the three alternative productivity series. The solid line uses the calibrated common productivity component, A_t, while the dashed orange line uses the industry-specific productivity series $A^{GDP}it$ that matches both relative industry prices and real per capita GDP growth. The dashed black line instead uses the industry-specific series $A^{BLS}it$, which matches relative price changes while aligning aggregate productivity growth with the BLS series. All three specifications generate a substantial increase in the skill premium over the sample period and produce very similar dynamics. Allowing for heterogeneous industry-specific productivity growth generates a somewhat larger increase in the relative wage than the common-productivity specification, while the BLS-aligned series produces a path that closely tracks the other two. This similarity suggests that the model's prediction for the rise in the skill premium is not sensitive to the particular measure of aggregate

[15] In both the model and the data, real per capita GDP is calculated using a Fisher chained price index.

[16] Industry-level prices are obtained from Hubmer's replication kit and are derived from annual chained-price indices for gross output at the summary level for 71 industries.

Figure 4: Model predictions



(a) TFP **(b)** W^H

Notes: Panel (a) compares the evolution of the calibrated TFP series with the TFP series reported by the BLS. The solid line reports the common productivity component, A_t, calibrated to match real per capita GDP growth, while the dashed line reports the value-added-weighted average of industry-specific productivity, A_{it}^{GDP}, calibrated to match both real per capita GDP growth and relative industry prices. The BLS series is FRED series MFPPBS. Panel (b) reports the model-implied evolution of the relative wage of high-skill workers under three alternative productivity specifications: the common productivity component A_t; industry-specific productivity A_{it}^{GDP}, calibrated to match real per capita GDP growth and relative industry prices; and industry-specific productivity A_{it}^{BLS}, calibrated to match relative industry prices while aligning aggregate productivity growth with the BLS series.

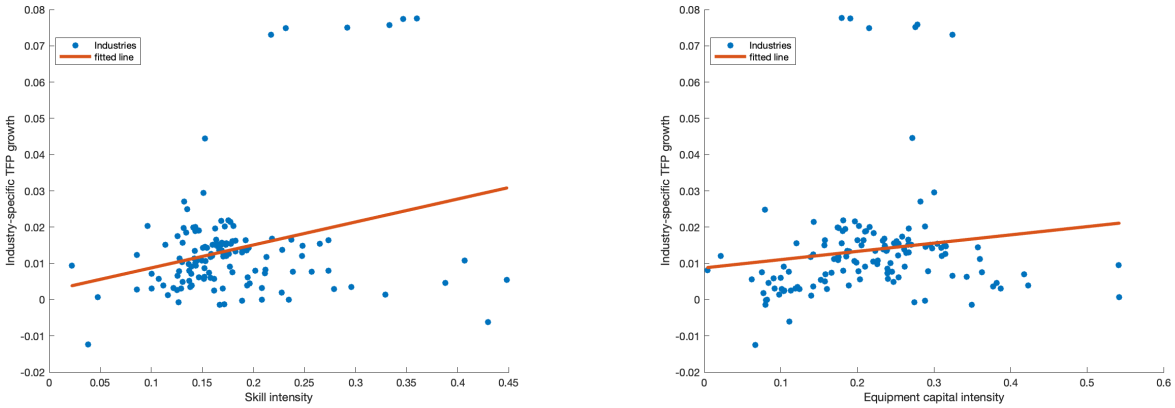
productivity growth used in the calibration.

In Figures 5a and 5b, I present the relationship between the calibrated industry-specific TFP growth rates ΔA_{it} and the factor intensities of high-skill workers and capital equipment in each industry. There is a positive correlation between industry-specific TFP growth and industry-level skill intensity, suggesting faster growth in the skill-intensive industries during this period. Moreover, there is also a weak positive correlation between ΔA_{it} and θ_i^E, suggesting a potentially positive interaction effect on the skill premium from the relative growth across industries and capital accumulation.

5 Quantitative analysis

In this section, I quantitatively study the contribution of different factors in driving changes in the skill premium using the calibrated model. I present two alternative versions of the

Figure 5: Correlation between TFP growth and factor intensities



(a) $Corr(\Delta A_{it}, \theta_i^H) = 0.270$ **(b)** $Corr(\Delta A_{it}, \theta_i^E) = 0.123$

Notes: ΔA_{it} denotes the total growth in calibrated productivity for industry i over 1982–2019. Skill intensities are calculated from OEWS-O*NET and capital equipment intensities are retrieved from Hubmer (2023). The fastest-growing industries are: 1) other measuring and controlling device manufacturing, 2) electronic computer manufacturing, 3) other communications equipment manufacturing, 4) telephone apparatus manufacturing, 5) audio and video equipment manufacturing, 6) magnetic and optical recording media manufacturing. In Appendix Table 8 and 9, I report average annual industry-level TFP growth for all 133 industries.

model with and without the consumption upgrading channel. Next, I use counterfactuals to investigate the effects of each individual shock on labor-market outcomes.

5.1 Sources of the Rising Skill Premium

The baseline model with nonhomothetic preferences and calibrated industry-specific productivity, A_{it}^{GDP}, predicts a 38.1% increase in the relative wage of high-skill workers over the sample period.[17] Figure 4b shows that the predicted path of the skill premium is similar across the three alternative productivity specifications. Using a common productivity component, denoted A_t, generates a somewhat smaller increase, while aligning industry productivity with the BLS TFP series produces an increase close to the baseline.

Nonhomothetic preferences amplify the rise in the skill premium across all three productivity specifications. As shown in Table 3, the homothetic model predicts increases

[17] For comparison, the college premium rises from 1.33 to 1.88 between 1977 and 2005 in the World KLEMS data used by Buera et al. (2022), an increase of about 41%, and from 1.48 to 1.89, or about 28%, in the composition-adjusted CPS series of Acemoglu and Autor (2011). Comin et al. (2026) report a 19.5 log point increase in the wage of high-skill relative to middle-skill occupations between 1980 and 2016.

Table 3: Model Comparison

	Homothetic ($\gamma_{it} = 1$)			Nonhomothetic		
	(1)	(2)	(3)	(4)	(5)	(6)
	A_t	A_{it}^{GDP}	A_{it}^{BLS}	A_t	A_{it}^{GDP}	A_{it}^{BLS}
$\Delta \frac{W_t^H}{W_t^L}$	30.3%	32.8%	33.1%	34.9%	38.1%	37.4%

Notes: This table reports the model-implied increase in the relative wage of high-skill workers under homothetic and nonhomothetic preferences and three alternative productivity specifications. Columns (1)–(3) report results under homothetic preferences, and columns (4)–(6) under nonhomothetic preferences. Columns (1) and (4) use common productivity, A_t, calibrated to match real per capita GDP growth (FRED series A939RX0Q048SBEA). Columns (2) and (5) use industry-specific productivity, A_{it}^{GDP}, calibrated to match both real per capita GDP growth and changes in relative industry prices. Columns (3) and (6) use A_{it}^{BLS}, constructed by rescaling the GDP-calibrated industry productivity paths so that the aggregate productivity index follows the BLS TFP series (FRED series MFPPBS), while preserving relative productivity across industries.

of 30.3%, 32.8%, and 33.1% under A_t, A_{it}^{GDP}, and A_{it}^{BLS}, respectively. The corresponding increases under nonhomothetic preferences are 34.9%, 38.1%, and 37.4%. Allowing consumption patterns to change systematically with income therefore raises the predicted increase in the skill premium by approximately 4.3–5.3 percentage points.

To identify the sources of this increase, I separate the calibrated industry productivity paths into aggregate and relative industry components. Suppressing the superscript GDP for the baseline paths, I define

$$A_t \equiv \frac{\sum_i \omega_{i0} A_{it}}{\sum_i \omega_{i0} A_{i0}}, \qquad \frac{\Delta A_{it}}{\Delta A_t} \equiv \frac{A_{it}/A_{i0}}{A_t/A_0}. \tag{22}$$

The fixed weights $\omega_{i0} = f\omega_{i0}^H + (1-f)\omega_{i0}^L$ are population-weighted averages of the two household groups' initial expenditure shares. The aggregate index is normalized to $A_0 = 1$. In the productivity-growth ratio, ΔA_{it} and ΔA_t denote gross growth factors relative to the initial period. Thus, $\Delta A_{it}/\Delta A_t > 1$ indicates that industry productivity grows faster than the aggregate index.

Table 4 reports counterfactuals for four forces: aggregate productivity growth, structures accumulation, equipment accumulation, and relative industry productivity growth. Each exercise allows one component to evolve while holding the others fixed and reports the resulting change in the skill premium under homothetic and nonhomothetic preferences.

I first isolate aggregate productivity growth, holding both capital stocks and rela-

Table 4: Decomposition of the Rising Skill Premium

	Homothetic		Nonhomothetic	
	$\Delta \frac{W_t^H}{W_t^L}$	Decomposition	$\Delta \frac{W_t^H}{W_t^L}$	Decomposition
Total	32.8%	100%	38.1%	100%
ΔA_t	0.0%	0.0%	3.2%	8.4%
ΔS_t	-0.66%	-2.0%	0.72%	1.9%
ΔK_t	30.9%	94.1%	31.8%	83.5%
$\Delta A_{it}/\Delta A_t$	0.72%	2.2%	0.15%	0.4%

Notes: This table reports counterfactual changes in the relative wage of high-skill workers under the baseline industry-specific productivity calibration, A_{it}^{GDP}. The first two columns report results under homothetic preferences and the last two under nonhomothetic preferences. Each counterfactual varies one component: aggregate productivity A_t, structures capital S_t, equipment capital K_t, or relative industry productivity $\Delta A_{it}/\Delta A_t$, while holding the other components fixed. The decomposition column reports each counterfactual change as a share of the total model-implied increase in the skill premium, calculated from unrounded estimates. These are separate nonlinear counterfactuals, so their shares need not sum to 100 percent.

tive industry productivity fixed. Under homothetic preferences, common productivity growth is skill neutral and has essentially no effect on the relative wage. Under nonhomothetic preferences, aggregate productivity growth raises real expenditure and shifts demand toward goods and services with higher expenditure elasticities. Because these industries are, on average, more skill intensive, this income-driven reallocation raises the relative demand for high-skill labor. Quantitatively, aggregate productivity growth increases the skill premium by 3.2%, accounting for 8.4% of the total increase in the nonhomothetic model.

Second, I isolate structures accumulation. Structures are skill neutral within an industry, but changes in their rental price affect industries differently because structures intensities vary. Under homothetic preferences, structures accumulation reduces the skill premium by 0.66%, reflecting a modest reallocation toward less skill-intensive industries. With nonhomothetic preferences, the associated increase in real income introduces an additional demand effect, and the net effect becomes positive: structures accumulation raises the skill premium by 0.72%, accounting for 1.9% of the total increase.

Third, equipment accumulation is quantitatively the dominant source of the rising skill premium. Capital–skill complementarity makes equipment accumulation favor high-skill relative to low-skill labor. In the homothetic model, equipment accumulation alone

increases the skill premium by 30.9%, accounting for 94.1% of the total predicted increase. Under nonhomothetic preferences, the increase is slightly larger, at 31.8%, but represents a smaller share of the total, 83.5%. This lower share reflects the additional contribution of other forces, particularly aggregate productivity growth operating through consumption upgrading.

Finally, relative industry productivity growth, $\Delta A_{it}/\Delta A_t$, changes relative goods prices and reallocates expenditure and production across industries. This channel has a small positive effect in both models: it raises the skill premium by 0.72% under homothetic preferences and 0.15% under nonhomothetic preferences, accounting for 2.2% and 0.4% of their respective total increases. Thus, despite the correlation between industry productivity growth and skill intensity, the isolated relative-productivity channel contributes little to the rising skill premium in the baseline decomposition.

Together, the isolated counterfactuals account for approximately 94% of the total increase in each model. The remaining difference reflects their joint effects in general equilibrium and cannot be assigned to a single channel by these exercises. The positive relationship between industry productivity growth and equipment intensity in Figure 5b suggests one potential interaction: productivity growth in equipment-intensive industries may reinforce the effects of equipment accumulation on relative labor demand.

I also repeat the decomposition using BLS-aligned productivity, the capital–skill complementarity elasticities estimated by Krusell et al. (2000), and Stone–Geary preferences. Appendix Tables 5, 6, and 7 show that equipment accumulation remains the dominant source of the rising skill premium, while nonhomothetic preferences generate a larger total increase than the homothetic benchmark. Some secondary effects are sensitive to the specification. In particular, relative industry productivity growth lowers the skill premium by 1.4% under Stone–Geary preferences, compared with a 0.15% increase in the baseline. The contribution of consumption upgrading is also smaller under the Krusell et al. (2000) elasticities, as discussed in Appendix 7.2.

5.2 Comparison with the Literature

My mechanism is closely related to Buera et al. (2022) and Comin et al. (2026), both of whom emphasize that changes in the composition of demand can affect relative labor demand when sectors differ in their skill intensity. In Buera et al. (2022), sector-specific skill-neutral technological change reallocates economic activity toward the skill-intensive

sector and thereby raises the relative demand for skilled labor. Their calibrated model matches the increase in the U.S. skill premium from 1.33 to 1.88 between 1977 and 2005. They find that sector-specific skill-neutral technological change, which combines productivity growth common across sectors with relative productivity growth across sectors, accounts for 18-24% of the increase in the skill premium attributable to technological change.

The income channel in Comin et al. (2026) is broader. Their model features three occupational groups and eight sectors, and nonhomothetic preferences imply that shocks affecting either the mean or the dispersion of household income reallocate expenditure toward sectors with higher expenditure elasticities. Because these sectors are relatively intensive in high- and low-skill occupations, this reallocation generates labor-market polarization. Between 1980 and 2016, the relative wage of high- to middle-skill workers rises by 19.5 log points in the data, of which their income channel accounts for 8.3 log points, or 42%. About one-quarter of this channel operates through changes in the dispersion of household income rather than changes in average income.

My framework differs from these papers along several important dimensions. First, I explicitly model the role of capital in generating changes in relative labor demand. Buera et al. (2022) abstract from capital and interpret changes in sectoral skill intensity as a reduced-form measure of skill-biased technological change. Comin et al. (2026) likewise abstract from capital as a production input and model sectoral output using three occupational labor inputs. In contrast, my production structure distinguishes equipment and structures capital and allows equipment to be differentially substitutable with high- and low-skill labor. I estimate these substitution elasticities directly and document evidence of capital–skill complementarity. Equipment accumulation therefore constitutes an explicit source of skill-biased technological change in my model rather than being absorbed into changes in labor factor intensities.

Second, the papers differ in how skill is defined and how changes in labor composition enter the analysis. Buera et al. (2022) define high-skill workers as those with a college degree or more, while Comin et al. (2026) follow Acemoglu and Autor (2011) in grouping occupations into high-, middle-, and low-skill categories based on their 1980 average hourly wages. I instead classify occupations using O*NET Job Zones, which capture the preparation and skill requirements of occupations rather than workers' educational attainment or realized occupational wages. Relative to these alternative definitions, my classification isolates a smaller group of highly skilled occupations whose aggregate em-

ployment share changes relatively little over time. I therefore hold the aggregate employment shares of high- and low-skill occupations fixed in the baseline quantitative exercise, allowing the analysis to focus on changes in relative labor demand. Comin et al. (2026) additionally allow changes in the income distribution to affect demand through household heterogeneity, which can further amplify the nonhomothetic income channel.

Third, the papers differ in how the input–output structure is incorporated. Both Buera et al. (2022) and Comin et al. (2026) organize sectoral demand in terms of value-added consumption: observed final expenditures are mapped through the input–output structure and attributed back to the industries in which value added is produced. In contrast, I organize the economy around final-demand categories and attribute to each category the labor and capital inputs used both directly and indirectly throughout its production chain. This corresponds to the *final expenditure approach* of Herrendorf et al. (2013). As a result, the skill intensity associated with a final consumption category reflects its full factor content rather than only the factor intensity of the industry in which value added is directly produced. In my data, this construction produces less dispersion in skill intensity across final-demand categories, making the expenditure-reallocation channel more conservative along this dimension.

6 Conclusion

Using detailed household consumption data together with occupation-by-industry employment data, I document that higher-income households devote a larger share of expenditure to skill-intensive goods and services, and that the consumption share of these goods and services has risen over time. These patterns imply that economic growth can raise the relative demand for skilled labor through consumption upgrading, providing a demand-side source of rising wage inequality in addition to conventional production-side explanations.

To quantify the relative importance of these forces, I develop a multi-industry general equilibrium model with two skill groups, nonhomothetic preferences, and capital–skill complementarity. The model distinguishes two conceptually different mechanisms. On the demand side, aggregate income growth reallocates expenditure toward goods and services with higher expenditure elasticities, which are also more skill intensive. On the production side, equipment accumulation raises the relative demand for skilled labor because equipment is more substitutable with low-skill labor and more complementary

to high-skill labor. Industry heterogeneity in factor intensities and productivity growth determines how strongly these aggregate forces translate into changes in relative labor demand.

Calibrated to U.S. data from 1982 to 2019, the model generates a 38.1% increase in the relative wage of high-skill workers. Capital–skill complementarity is quantitatively the dominant force: equipment accumulation accounts for over 80 percent of the predicted increase in the skill premium. Consumption upgrading nevertheless provides a meaningful additional source of skill demand. In particular, skill-neutral aggregate productivity growth has essentially no effect on the skill premium under homothetic preferences, but raises it under nonhomothetic preferences by shifting expenditure toward more skill-intensive goods and services. Thus, consumption upgrading amplifies the rise in wage inequality generated by production-side technological change, while capital–skill complementarity remains the primary quantitative driver.

The comparison between these two channels also highlights the importance of modeling both production and demand in a disaggregated economy. Capital–skill complementarity operates primarily through changes in relative factor demand within industries, whereas consumption upgrading operates through the reallocation of expenditure across industries with different skill intensities. Accounting for both mechanisms provides a more complete interpretation of the rise in the skill premium and clarifies why skill-neutral economic growth can become skill-biased in general equilibrium.

An important direction for future work is to endogenize labor supply and occupational choice in order to study the welfare and distributional consequences of these mechanisms. In particular, incorporating consumption upgrading into models of optimal taxation may affect the trade-off between redistribution and efficiency because changes in household income alter not only labor supply but also the composition of demand and, through it, the relative demand for different types of workers.

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7 Appendix A: Additional tables and figures

7.1 Main results using BLS TFP series

I redo the quantitative exercises using A_{it}^{BLS}. The results, displayed in Table 5, are similar to those from my baseline analysis. Because my calibrated aggregate TFP growth is slightly higher than the BLS estimates, the impact on the relative wage increase from A_t is marginally larger in my baseline model.

Table 5: Decomposition using BLS TFP

	Homothetic		Nonhomothetic	
	$\Delta \frac{W_t^H}{W_t^L}$	Decomposition	$\Delta \frac{W_t^H}{W_t^L}$	Decomposition
Total	33.1%	100%	37.4%	100%
ΔA_t	0%	0%	2.4%	6.4%
ΔS_t	-0.66%	-2.0%	0.7%	1.9%
ΔK_t	30.9%	93.4%	31.8%	85.0%
$\Delta A_{it}/\Delta A_t$	0.9%	2.8%	0.15%	0.4%

Notes: This table decomposes the model-implied increase in the relative wage of high-skill workers using the BLS-aligned industry-specific productivity series, A_{it}^{BLS}. The first two columns report results under homothetic preferences, and the last two columns under nonhomothetic preferences. Each counterfactual varies one exogenous force—aggregate productivity A_t, structures capital S_t, equipment capital K_t, or relative industry productivity $\Delta A_{it}/\Delta A_t$—while holding the remaining forces fixed at their initial values. The decomposition column reports each counterfactual change as a share of the total model-implied increase in the skill premium, computed from unrounded values. Because the model is nonlinear, the individual contributions need not sum to 100 percent.

7.2 Main results under capital–skill complementarity elasticities from Krusell et al. (2000)

I repeat the quantitative exercises using the substitution elasticities estimated by Krusell et al. (2000), setting $\rho = 1.67$ and $\eta = 0.67$ in my nesting structure.[18] This exercise is a test of sensitivity to the magnitudes of the elasticities.

Both alternative elasticities exceed my baseline estimates of $\rho = 1.23$ and $\eta = 0.41$, and the change has two opposing effects. On the one hand, the gap between the two elasticities is larger, $\rho - \eta = 1.00$ compared with 0.82 in the baseline, which strengthens

[18] Krusell et al. (2000) estimate these elasticities under a different nesting, in which equipment is nested with high-skill labor.

capital–skill complementarity and raises the relative demand for high-skill labor generated by the same increase in equipment. On the other hand, the higher value of η makes high- and low-skill labor more substitutable in my nesting, so any given shift in relative skill demand produces a smaller change in the relative wage.

Table 6 reports the resulting decomposition. Under nonhomothetic preferences, the full model predicts a 31.6% increase in the relative wage of high-skill workers, compared with 38.1% under my baseline elasticities. The equipment-only counterfactual generates a 27.2% increase, compared with 31.8% in the baseline. Thus, the dampening effect of greater substitutability between the two labor types outweighs the stronger capital–skill complementarity. The same dampening reduces the consumption-upgrading channel: aggregate productivity growth alone raises the skill premium by 1.4%, compared with 3.2% in the baseline, because the reallocation of expenditure toward skill-intensive goods and services translates into a smaller wage response.

As a share of the full-model increase, equipment accumulation accounts for 86.0% of the total change, compared with 83.5% in the baseline. This higher share does not reflect a larger absolute effect of equipment accumulation; rather, the full-model increase falls proportionally more than the equipment-only increase. Aggregate productivity growth accounts for 4.4% of the total, compared with 8.4% in the baseline. Overall, equipment accumulation remains the dominant quantitative force under the alternative elasticities. The contribution of consumption upgrading is more sensitive to the elasticities, and my baseline estimates imply a larger role for it than the Krusell et al. (2000) elasticities do.

Table 6: Decomposition Using Capital–Skill Complementarity Elasticities from Krusell et al. (2000)

	Homothetic		Nonhomothetic	
	$\Delta \frac{W_t^H}{W_t^L}$	Decomposition	$\Delta \frac{W_t^H}{W_t^L}$	Decomposition
Total	29.0%	100%	31.6%	100%
ΔA_t	0%	0%	1.4%	4.4%
ΔS_t	-0.2%	-0.8%	0.4%	1.3%
ΔK_t	26.5%	91.5%	27.2%	86.0%
$\Delta A_{it}/\Delta A_t$	1.5%	5.3%	1.2%	3.8%

Notes: This table decomposes the model-implied increase in the relative wage of high-skill workers using the substitution elasticities estimated by Krusell et al. (2000), mapped to $\rho = 1.67$ and $\eta = 0.67$ in my nesting structure. The first two columns report results under homothetic preferences, and the last two columns under nonhomothetic preferences. All specifications use the calibrated industry-specific productivity series, A_{it}^{GDP}. Each counterfactual varies one exogenous force—aggregate productivity A_t, structures capital S_t, equipment capital K_t, or relative industry productivity $\Delta A_{it}/\Delta A_t$—while holding the remaining forces fixed at their initial values. The decomposition column reports each counterfactual change as a share of the total model-implied increase in the skill premium, computed from unrounded values. Because the model is nonlinear, the individual contributions need not sum to 100 percent.

7.3 Main results using Stone–Geary preferences

Table 7 reports the quantitative results under Stone–Geary preferences. The model predicts a 39.9% increase in the relative wage of high-skill workers, compared with 38.1% under the baseline demand system and 32.8% in the common homothetic benchmark. Equipment accumulation remains the dominant source of the rising skill premium: the equipment-only counterfactual generates a 33.8% increase, equivalent to 84.8% of the full-model increase. Aggregate productivity growth alone generates a 3.3% increase, equivalent to 8.3% of the total. The contribution of relative industry productivity changes, however, reverses sign, from a 0.15% increase under the baseline demand system to a 1.4% decline under Stone–Geary preferences. Thus, the dominance of equipment accumulation and the larger total increase relative to the homothetic benchmark remain robust, although the sign of the relative-productivity effect is sensitive to the demand specification.

Table 7: Decomposition under Stone–Geary Preferences

	Homothetic		Nonhomothetic	
	$\Delta \frac{W_t^H}{W_t^L}$	Decomposition	$\Delta \frac{W_t^H}{W_t^L}$	Decomposition
Total	32.8%	100%	39.9%	100%
ΔA_t	0.0%	0.0%	3.3%	8.3%
ΔS_t	-0.66%	-2.0%	1.6%	4.0%
ΔK_t	30.9%	94.1%	33.8%	84.8%
$\Delta A_{it}/\Delta A_t$	0.72%	2.2%	-1.4%	-3.4%

Notes: This table reports counterfactual changes in the relative wage of high-skill workers using the estimated production elasticities $\rho = 1.2341$ and $\eta = 0.4144$, and the consumption substitution elasticity $\sigma = 1.55$. The nonhomothetic column uses Stone–Geary preferences calibrated to the initial equilibrium, with preference parameters held fixed across counterfactuals and productivity paths taken from the baseline nonhomothetic calibration. The homothetic column reproduces the baseline homothetic benchmark in Table 4, using that benchmark's calibration and counterfactual definitions. Relative industry productivity denotes $\Delta A_{it}/\Delta A_t$. The decomposition column reports each counterfactual change as a share of the total model-implied increase, calculated from unrounded estimates. These are separate nonlinear counterfactuals, so their shares need not sum to 100 percent.

7.4 Calibration of the Stone–Geary preference parameters

This section describes the calibration of the Stone–Geary preference parameters used in the robustness exercise. Following Sancho (2024), I assume that households share a common set of preference weights, $\{a_i\}_{i\in I}$, and subsistence consumption parameters, $\{\bar{z}_i\}_{i\in I}$. The elasticity of substitution across goods, σ, is taken from Hubmer (2023).

Under Stone–Geary preferences, demand for good i is given by

$$C_{it} = \bar{z}_i + \frac{s_i(\mathbf{P}_t)}{P_{it}} \left(E_t - \sum_{j\in I} P_{jt}\bar{z}_j \right), \tag{23}$$

where E_t denotes total expenditure and

$$s_i(\mathbf{P}_t) = \frac{a_i^\sigma P_{it}^{1-\sigma}}{\sum_{j\in I} a_j^\sigma P_{jt}^{1-\sigma}} \tag{24}$$

is good i's share of expenditure net of subsistence expenditure.

Let t_0 denote the initial period. Because all initial-period prices are normalized to one,

equation (24) becomes

$$s_i(\mathbf{1}) = \frac{a_i^\sigma}{\sum_{j \in I} a_j^\sigma}. \tag{25}$$

The expenditure elasticity of demand for good i is

$$\begin{aligned} \gamma_{it} &= \frac{\partial C_{it}}{\partial E_t} \frac{E_t}{C_{it}} \\ &= \frac{s_i(\mathbf{P}_t)}{P_{it}} \frac{E_t}{C_{it}} = \frac{s_i(\mathbf{P}_t)}{\omega_{it}}, \end{aligned} \tag{26}$$

where

$$\omega_{it} = \frac{P_{it} C_{it}}{E_t}$$

is the expenditure share of good i. It follows that, in the initial period,

$$s_i(\mathbf{1}) = \gamma_{it_0} \omega_{it_0}. \tag{27}$$

Given the expenditure elasticities estimated by Hubmer (2023) and the initial expenditure shares observed in the data, equation (27) identifies $s_i(\mathbf{1})$. Combining this result with equation (25) identifies the relative preference weights a_i. I impose the normalization

$$\sum_{i \in I} a_i = 1 \tag{28}$$

to determine their levels.

I calibrate the subsistence parameters using the Frisch money-flexibility parameter, denoted by ϕ. Following Frisch (1959), I set $\phi = -2$. With initial-period prices normalized to one, the money-flexibility parameter satisfies

$$-\phi = \frac{E_{t_0}}{E_{t_0} - \sum_{j \in I} \bar{z}_j}. \tag{29}$$

Therefore, initial supernumerary expenditure is

$$E_{t_0} - \sum_{j \in I} \bar{z}_j = \frac{E_{t_0}}{-\phi}. \tag{30}$$

Substituting equation (30) into the initial-period demand system gives the good-specific subsistence parameters:

$$\bar{z}_i = C_{it_0} - s_i(\mathbf{1}) \frac{E_{t_0}}{-\phi}. \tag{31}$$

The calibration uses initial-period consumption shares and expenditure elasticities from the data, together with prices, consumption quantities, and aggregate expenditure from the initial equilibrium of the baseline model. The calibrated parameters a_i and $\bar{z}_i$ are then held fixed throughout the quantitative exercise.

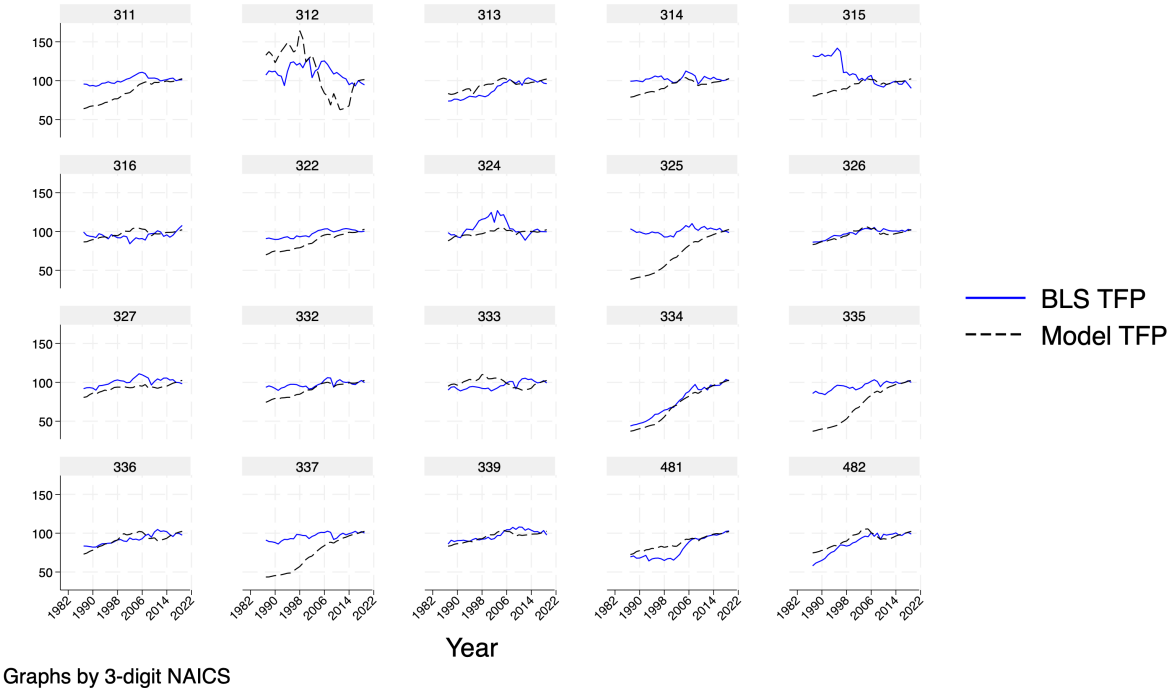
8 Appendix B: Industry-specific TFP growth

8.1 Comparison with BLS TFP for selected industries

The BLS reports total factor productivity (TFP) estimates for several major industries, primarily in manufacturing, beginning in 1987.[19] Figure 6 compares the calibrated industry-level TFP series with the corresponding BLS estimates for 20 industries at the three-digit NAICS level. I aggregate the calibrated series to the three-digit level by taking the unweighted average across the final-demand industries within each NAICS group. Overall, the calibrated series follow trends similar to the BLS estimates for most of the industries shown.

[19] Source: U.S. Bureau of Labor Statistics, https://www.bls.gov/productivity/tables/.

Figure 6: Model-Calibrated and BLS-Estimated Industry TFP Growth



8.2 List of industries

Tables 8 and 9 report the average annual TFP growth rates for the 133 final-demand industries in the model over 1982–2019. For each industry, annual TFP growth is calculated as the log change in calibrated TFP from the previous year and then averaged over the sample period. The industry-level TFP series are calibrated to match changes in relative industry prices, while the common productivity component is chosen so that the model matches the growth of real per capita GDP, as described in Section 4.

Table 8: Average Annual TFP Growth by Industry

BEA I-O code	Industry Title	Avg. annual $\Delta \ln A_{it}$
oohre	OOH + Real Estate	-0.0134
541100	Legal services	-0.0090
52A000	Monetary authorities and depository credit intermediation	-0.0034
623000	Nursing and residential care facilities	-0.0033
512100	Motion picture and video industries	-0.0024
522A00	Nondepository credit intermediation and related activities	-0.0024
712000	Museums, historical sites, zoos, and parks	-0.0021
622000	Hospitals	-0.0021
611B00	Other educational services	-0.0011
812900	Other personal services	-0.0004
711200	Spectator sports	-0.0002
3122A0	Tobacco product manufacturing	0.0004
621B00	Medical and diagnostic labs and outpatient and other ambulatory care services	0.0005
624400	Child day care services	0.0006
621A00	Offices of physicians, dentists, and other health practitioners	0.0010
7211A0	Hotels and motels, including casino hotels	0.0010
621600	Home health care services	0.0011
23	Construction	0.0011
812100	Personal care services	0.0013
325610	Soap and cleaning compound manufacturing	0.0014
812300	Dry-cleaning and laundry services	0.0015
811400	Personal and household goods repair and maintenance	0.0016
524100	Insurance carriers	0.0018
325620	Toilet preparation manufacturing	0.0019
611A00	Junior colleges, colleges, universities, and professional schools	0.0019
722000	Food services and drinking places	0.0022
325320	Pesticide and other agricultural chemical manufacturing	0.0023
325412	Pharmaceutical preparation manufacturing	0.0024
611100	Elementary and secondary schools	0.0027
221200	Natural gas distribution	0.0033
713A00	Amusement parks, arcades, and gambling industries	0.0034
48A000	Scenic and sightseeing transportation and support activities for transportation	0.0037
221100	Electric power generation, transmission, and distribution	0.0039
325510	Paint and coating manufacturing	0.0041
8111A0	Automotive repair and maintenance, except car washes	0.0041
492000	Couriers and messengers	0.0041
812200	Death care services	0.0043
532100	Automotive equipment rental and leasing	0.0045
532230	Video tape and disc rental	0.0049
3259A0	All other chemical product and preparation manufacturing	0.0051
561700	Services to buildings and dwellings	0.0053
813B00	Civic, social, professional, and similar organizations	0.0053
811200	Electronic and precision equipment repair and maintenance	0.0054
532400	Commercial and industrial machinery and equipment rental and leasing	0.0055
541940	Veterinary services	0.0055
562000	Waste management and remediation services	0.0056
561900	Other support services	0.0057
322291	Sanitary paper product manufacturing	0.0059
311000	Food at Home	0.0061
312100	Alcohol	0.0062
485000	Transit and ground passenger transportation	0.0063
532A00	General and consumer goods rental except video tapes and discs	0.0067
221300	Water, sewage, and other systems	0.0071
32712A	Brick, tile, and other structural clay product manufacturing	0.0074
561600	Investigation and security services	0.0075
541200	Accounting, tax preparation, bookkeeping, and payroll services	0.0078
337910	Mattress manufacturing	0.0081
322230	Stationery product manufacturing	0.0083
337122	Nonupholstered wood household furniture manufacturing	0.0087
337212	Office furniture and custom architectural woodwork and millwork manufacturing	0.0087
33712A	Metal and other household furniture (except wood) manufacturing	0.0089
327212	Other pressed and blown glass and glassware manufacturing	0.0089

Table 9: Average Annual TFP Growth by Industry, Continued

BEA I-O code	Industry Title	Avg. annual $\Delta \ln A_{it}$
337121	Upholstered household furniture manufacturing	0.0096
33221A	Cutlery, utensil, pot, and pan manufacturing	0.0097
483000	Water transportation	0.0097
482000	Rail transportation	0.0097
515200	Cable and other subscription programming	0.0097
337920	Blind and shade manufacturing	0.0099
481000	Air transportation	0.0099
32711A	Pottery, ceramics, and plumbing fixture manufacturing	0.0100
332913	Plumbing fixture fitting and trim manufacturing	0.0101
333991	Power-driven handtool manufacturing	0.0105
337110	Wood kitchen cabinet and countertop manufacturing	0.0106
336991	Motorcycle, bicycle, and parts manufacturing	0.0108
491000	Postal service	0.0109
333415	Air conditioning, refrigeration, and warm air heating equipment manufacturing	0.0110
333112	Lawn and garden equipment manufacturing	0.0110
336612	Boat building	0.0112
333315	Photographic and photocopying equipment manufacturing	0.0114
486000	Pipeline transportation	0.0114
33221B	Handtool manufacturing	0.0116
32619A	Other plastics product manufacturing	0.0117
212100	Coal mining	0.0117
333618	Other engine equipment manufacturing	0.0118
511130	Book publishers	0.0121
339930	Doll, toy, and game manufacturing	0.0123
333319	Other commercial and service industry machinery manufacturing	0.0124
335228	Other major household appliance manufacturing	0.0125
339113	Surgical appliance and supplies manufacturing	0.0127
335210	Small electrical appliance manufacturing	0.0127
33331A	Vending, commercial, industrial, and office machinery manufacturing	0.0130
326210	Tire manufacturing	0.0131
339940	Office supplies (except paper) manufacturing	0.0131
335120	Lighting fixture manufacturing	0.0132
335221	Household cooking appliance manufacturing	0.0133
339112	Surgical and medical instrument manufacturing	0.0134
335222	Household refrigerator and home freezer manufacturing	0.0135
339115	Ophthalmic goods manufacturing	0.0136
484000	Truck transportation	0.0136
335224	Household laundry equipment manufacturing	0.0137
33329A	Other industrial machinery manufacturing	0.0138
4A0000	Retail trade	0.0140
339920	Sporting and athletic goods manufacturing	0.0140
511120	Periodical publishers	0.0140
33999A	All other miscellaneous manufacturing	0.0142
339992	Musical instrument manufacturing	0.0143
339910	Jewelry and silverware manufacturing	0.0145
511110	Newspaper publishers	0.0146
314110	Carpet and rug mills	0.0158
314120	Curtain and linen mills	0.0167
315200	Apparel	0.0170
313100	Fiber, yarn, and thread mills	0.0172
316900	Other leather and allied product manufacturing	0.0172
313210	Broadwoven fabric mills	0.0176
336110	Car and Other Vehicles	0.0181
315100	Apparel knitting mills	0.0181
316200	Footwear manufacturing	0.0181
517000	Telecommunications	0.0182
420000	Wholesale trade	0.0182
111400	Greenhouse, nursery, and floriculture production	0.0188
315900	Apparel accessories and other apparel manufacturing	0.0195
336300	Motor vehicle parts manufacturing	0.0197
336214	Travel trailer and camper manufacturing	0.0198
493000	Warehousing and storage	0.0232
324110	Petroleum refineries	0.0254
324191	Petroleum lubricating oil and grease manufacturing	0.0277
211000	Oil and gas extraction	0.0427
334613	Magnetic and optical recording media manufacturing	0.0705
334210	Telephone apparatus manufacturing	0.0720
334300	Audio and video equipment manufacturing	0.0723
334111	Electronic computer manufacturing	0.0725
334290	Other communications equipment manufacturing	0.0743
33451A	Other Measuring and Controlling Device Manufacturing	0.0745